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Enhancing Long-Form Video Question Answering via CLIP-driven Semantic Frame Retrieval Pipeline

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Declaration

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Abstract

Video Large Language Models (Vid-LLMs) are limited by a fixed context window, which prevents them from processing the full duration of long-form videos. The most common response is using uniform sampling to select frames at fixed temporal intervals, regardless of the question being asked. This project investigates whether question-aware semantic retrieval outperforms uniform sampling in multiple-choice question-answering accuracy on long-form videos.

The pipeline is designed to be training-free, single-pass, and question-aware. Video frames are encoded using CLIP and stored in a FAISS Index. At query time, the concatenated question and answer options are encoded as a text embedding and used to retrieve the $k = 5$ most similar frames. These frames are passed to LLaVA-Next 13B for downstream question-answering. This pipeline is evaluated on the Video-MME benchmark, covering all domains, duration tiers, and sub-categories.

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Chapter 1

Introduction

1.1 Motivation

The rapid rise of online video has created a fundamental challenge for modern AI. Video Large Language Models (Vid-LLMs) are powerful multi-modal tools that demonstrate impressive capabilities in analysing visual content to answer user queries. However, the context window of even the most capable Vid-LLMs cannot handle the sheer volume of data in long-form content, as encoding every sampled frame into a sequence of visual tokens is computationally intractable.

For the purposes of this project, a "long video" is defined as a video ranging from 30 to 60 minutes in duration, following the definition from the Video-MME benchmark [12]. Take a 45-minute video sampled at 1 frame per second, yielding 2,700 sampled frames. A standard ViT (Vision Transformer) uses 196 tokens per frame to encode visual data as a sequence of embedding vectors, producing 529,200 tokens. Modern open-source language models such as LLaVA-Next, trained with a context window of 4,096 tokens, cannot ingest input at this scale. Furthermore, the self-attention mechanism at the core of transformer architectures exhibits a complexity of $O(N^2)$, which scales quadratically with sequence length, leading to high computational and memory costs as token size increases[41]. Even with current hardware, processing the entire length of a moderately long video is computationally inefficient and not viable.

Currently, the industry standard for this challenge is uniform sampling: selecting a fixed number of frames at evenly spaced intervals across the video. Although straightforward to implement, the sampled frames are selected without any relation to the query, leading to a loss of fine-grained details [37]. For short and visually homogeneous videos, this approach is generally adequate. However, for long videos, uniform sampling frequently discards the frames needed to answer a given question, limiting the downstream accuracy of the language model [36]. This motivates an approach that selects the most semantically relevant frames based on the question itself rather than on temporal position, laying the foundation for the retrieval-augmented pipeline proposed in this project.

1.2 Problem Statement

This project asks a critical question: given a long-form video and a natural-language query, can a subset of semantically relevant frames be selected for that query in a computationally tractable manner?

This differs from general video understanding tasks such as video captioning or video summarisation. In this case, the selection criterion is linked to the semantics of the user query, with the main goal of question answering. Developing a pipeline to select semantically relevant frames will provide the downstream language model with stronger visual evidence, leading to improved question-answering accuracy. This project explores whether that hypothesis holds in practice by implementing and evaluating a retrieval-augmented pipeline against a uniform-sampling baseline on a standardised benchmark.

This directly addresses the core research question of the project:

Does semantic frame retrieval improve multiple-choice question-answering accuracy on long-form videos compared to uniform sampling, and if so, under which conditions (question type, video domain, and video duration)?

1.3 Aims and Objectives

The main aim of this project is to design, implement, and evaluate a training-free, retrieval-augmented pipeline to determine whether semantic retrieval improves downstream accuracy relative to a uniform sampling baseline.

Each objective below collectively operates together to answer the research question:

1. Formulate an evaluation framework using the Video-MME dataset[12], covering a range of video domains, durations, and question types. This is required to compare the uniform sampling baseline with semantic retrieval.
2. Design and build a modular RAG pipeline that uses CLIP[30] to encode frames and FAISS to index and retrieve the k most semantically relevant frames for a given text query.
3. Implement a uniform sampling baseline to serve as a point of comparison.
4. Prompt LLaVA-Next 13B[20] with the retrieved frames to generate multiple-choice answers, and measure and log accuracy across different retrieval methods.
5. Analyse logged results by question type and video domain to determine the conditions under which semantic retrieval outperforms uniform sampling.

1.4 Relationship to Program

This project links to several areas of the Computer Science degree program at Sheffield. The pipeline design is based on machine learning and mathematics, particularly transformer-based model architectures and contrastive learning. The retrieval system applies concepts from information retrieval and database systems, specifically similarity searches and these principles were covered in modules such as Text Processing. The evaluation methodology draws on empirical analysis and experimental design, as covered in modules such as Bio-Inspired Computing and Data-Driven Computing.

A specific challenge from this project is the computational scale. Running a 13-billion-parameter language model for visual inference requires powerful GPU hardware far beyond what is available on standard laptops. This required working with Sheffield's Stanage HPC cluster, introducing constraints around SLURM job scheduling and batch evaluation design that would not arise in typical coursework or assignments. The A100 40GB GPU enables experiments that would otherwise not be feasible on a standard laptop. Working with the Stanage infrastructure provided practical experience with HPCs that is relevant to research roles in machine learning.

1.5 Overview of the Dissertation

The following chapters of this project are structured as follows.

- **Chapter 2** presents an in-depth review covering the relevant literature, exploring the evolution of video understanding, examining the context bottleneck challenge in detail, and surveying current approaches, including techniques such as temporal compression, uniform sampling, and key-frame selection.
- **Chapter 3** explains the pipeline design, justifying key features and detailing the implementation of each aspect, and documents the technical challenges encountered throughout the process.
- **Chapter 4** describes the benchmark dataset, experimental conditions, and evaluation design used to compare the retrieval strategies.
- **Chapter 5** presents and analyses the experimental results from comparing semantic retrieval and uniform sampling, discussing in detail the link between the benchmark design and results.
- **Chapter 6** is the final chapter and concludes the dissertation and identifies any direction for future work.

Chapter 2

Literature Review

2.1 Overview

This chapter reviews the technical literature and prior approaches that motivate each design choice of the proposed retrieval-augmented pipeline, and identifies the gap in existing work. The chapter is structured as follows:

- Section 2.2 explores the evolution of video understanding from conventional methods to Video Large Language Models (Vid-LLMs), discussing why the context window has become the dominant challenge in long-form video analysis.
- Section 2.3 investigates the context-problem in detail, explaining the theory behind the token explosion and why uniform frame sampling is an insufficient response.
- Section 2.4 reviews five existing approaches for mitigating the context-problem, split into question-agnostic compression and question-aware selection approaches, and identifies the research gap this project addresses.
- Section 2.5 justifies the selection of CLIP as the retrieval encoder, exploring contrastive learning, the shared embedding space, and known limitations for video retrieval.
- Section 2.6 explores and justifies why FAISS was chosen as the vector indexing and similarity search mechanism.
- Section 2.7 explores the use of LLMs for downstream reasoning and describes how they are used for question-answering.
- Section 2.8 synthesises the identified research gap and summarises the chapter.

2.2 Evolution of Video Understanding

Video understanding has evolved through four main stages, each stage surpassing the limitations of its predecessor (see Figure 2.1). Understanding this evolution clarifies why Vid-LLMs are the current standard and explains why the context window has become the main challenge in long-form video analysis.

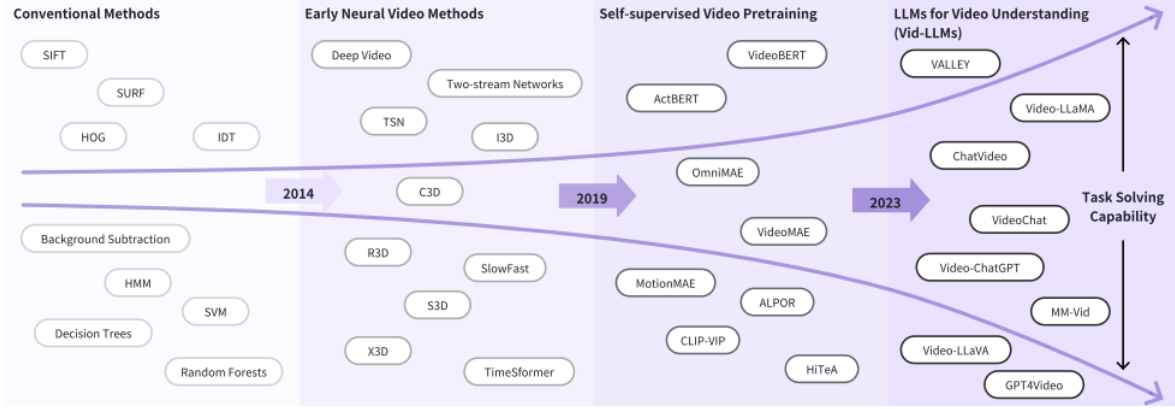


Figure 2.1: Evolution of Video Understanding [37]

2.2.1 Conventional Methods

Early video understanding methods relied on handcrafted feature extraction techniques to capture key visual information from individual frames. Commonly used descriptors included SIFT [23], SURF [3], and HOG [7], which encoded images into compact representations. To track temporal information, techniques such as optical flow [40] and Improved Dense Trajectories [42] have been used to track motion across frames.

Temporal analysis techniques, such as Hidden Markov Models [24], were used to understand the sequential structure of video events. Classification and recognition tasks used standard machine learning algorithms such as Support Vector Machines (SVMs) [32] and Decision Trees [50].

However, these early methods exhibited collective limitations: they relied on hand-crafted features, resulting in models that were less adaptable to unseen data, and performance gains required feature redesign rather than simply scaling training data. These collective limitations motivated the shift to learned representations through deep learning [37].

2.2.2 Deep Learning Approaches

Deep learning demonstrated superior task-solving capabilities compared to conventional methods by using deep neural networks, such as Convolutional Neural Networks (CNNs) [15], which

can learn features directly from raw data. However, even the best handcrafted feature methods outperformed neural networks such as 2D CNNs, which struggled to capture motion information due to the inadequate use of temporal cues [43]. Advancements in Two-Stream Networks [11], and 3D CNNs (e.g. C3D [39], I3D [5]) enabled learning from spatial, temporal, and spatio-temporal fields, demonstrating that modelling spatial and temporal features together drastically improved recognition accuracy (see Figure 2.2).

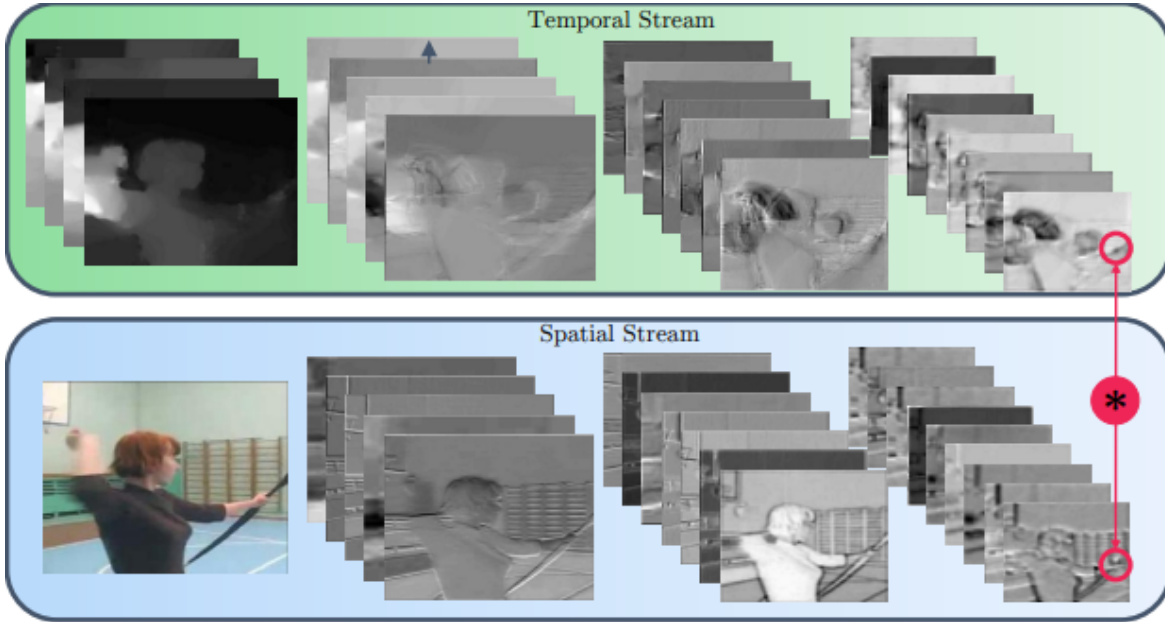


Figure 2.2: Two-Stream Network: Capturing spatial and temporal information [11]

The availability of large datasets (e.g., UCF-101[34], HMDB51[17], and Kinetics[16]) was necessary to train these models and achieve greater performance. For long-form videos, LSTM (Long Short-Term Memory) [28] and TSN (Temporal Segment Network)[43] were used to model sequential events.

The use of deep learning and neural methods reduced the dependence on hand-crafted features but still required large amounts of labelled data, introducing a new challenge for the field. This dependence on labelled data motivated the self-supervised pre-training paradigm discussed in the following subsection.

2.2.3 Self-Supervised Video Pre-training

The self-supervised pre-training paradigm enabled models to learn from unlabelled data, allowing them to generalise across diverse datasets. Rather than relying on manual labels, self-supervised techniques define pretext tasks from the video itself.

VideoBERT [35] was an early example that applied the BERT [8] language model. This approach tokenised video features, allowing the pretrained model to be fine-tuned for various

applications such as classification and video captioning. Following this, several other video masked auto-encoders such as VideoMAE[38], OmniMAE[13] and various training strategies (e.g., ActBERT [52], VLM [47]) reconstructed masked video patches to learn spatio-temporal features without supervision, as illustrated in Figure 2.3.

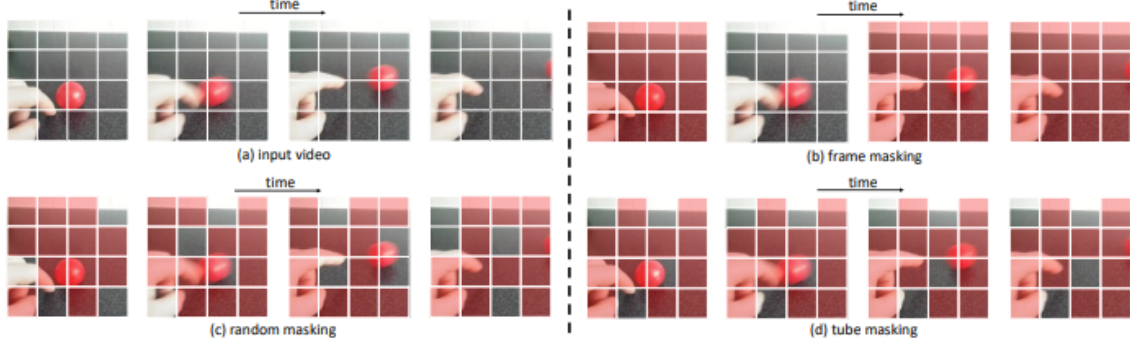


Figure 2.3: Four examples of frame masking [38]

While these advancements improved learning and data efficiency, these encoders produced fixed representations of video context but are not capable of complex, open-ended reasoning. Integrating a pre-trained LLM as the reasoning backbone enables open-ended reasoning and causal inference that encoder-only architectures cannot achieve, driving the Vid-LLM paradigm.

2.2.4 Video Large Language Models

The current generation of video understanding systems is defined by Vid-LLMs, combining visual encoders with the reasoning capabilities of large language models. This integration addresses the limitations of previous methods by enabling more advanced multi-modal understanding and human-like reasoning over videos [37]. The three primary frameworks are Video Analyser + LLM, Video Embedder + LLM, and Analyser + Embedder + LLM, as shown in Figure 2.4. In recent years, LLMs have advanced rapidly, with models such as GPT-4 [29] and Gemini 2.5 [6] trained on extensive datasets and equipped with in-context learning capabilities [4]. Prominent systems such as Video-LLaMA [51], Video-ChatGPT [26], and Video-LLaVA, [22] align visual features with LLM’s text embedding space, allowing the model to understand video content. These systems have demonstrated impressive performance on broader tasks, acting as general problem solvers and even solving specific user-requested tasks [37]. Understanding the architecture behind how Vid-LLMs process video is necessary to grasp why the context window has become the dominant challenge in the field, and is discussed in Section 2.2.5.

Despite this, Vid-LLMs face several challenges in real-world video understanding, such as the context window limitation of LLMs. Processing long videos requires converting thousands of frames into visual tokens. This exceeds the input capacity of most LLMs [45] [44], and

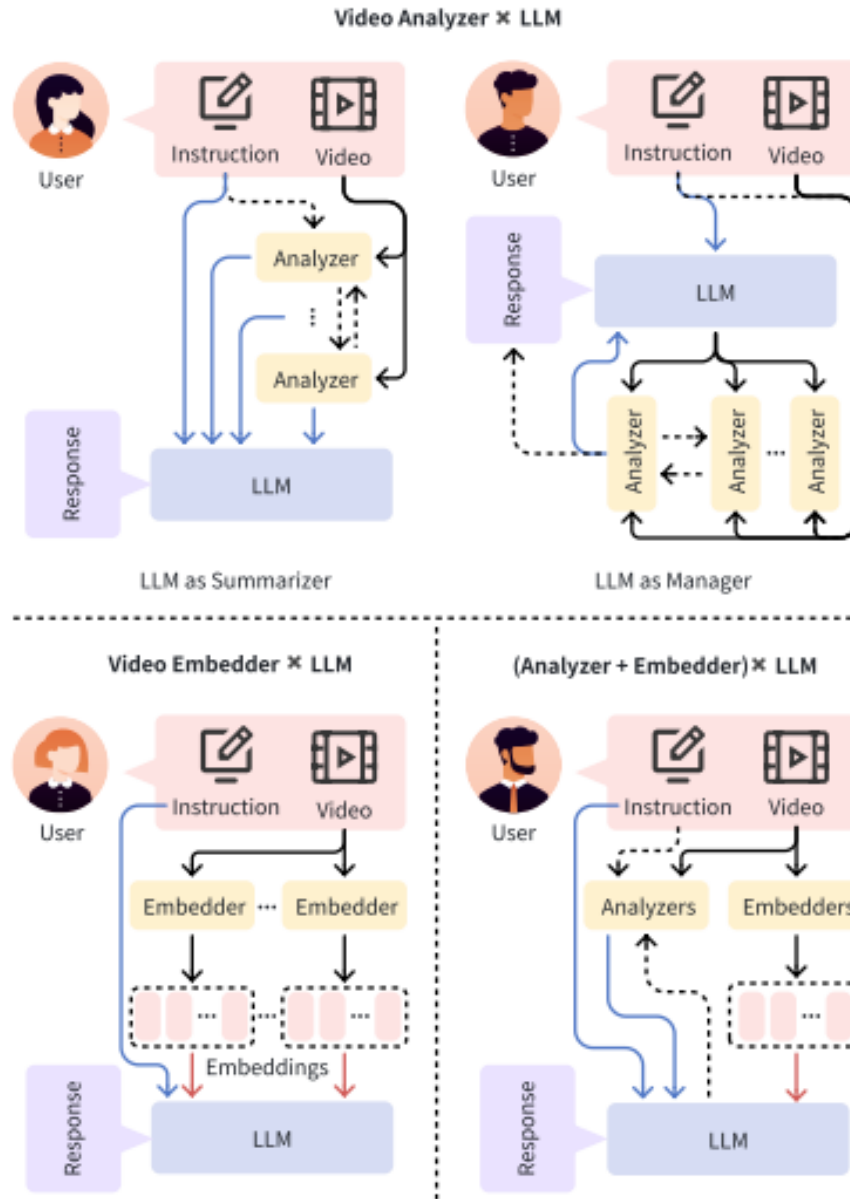


Figure 2.4: Three primary frameworks for Vid-LLMs [37]

forces Vid-LLMs to down-sample videos to a smaller number of frames, resulting in loss of information and, therefore, context. In addition, computing long sequences is computationally expensive and prevents real-time interaction. Addressing these challenges is the main aim for the retrieval-augmented approaches discussed in Section 2.3.

2.2.5 Vid-LLM Architecture

A standard Vid-LLM contains three components: a visual encoder, a projection module, and a large language model (see Figure 2.4).

The visual encoder is typically a Vision Transformer (ViT) [9] that transforms raw video frames into a mathematical representation for downstream processing. Given a video input of shape $T \times H \times W$, where T is the number of frames and H, W are spatial dimensions, each frame is first divided into fixed-size patches of shape $P \times P$, resulting in:

$$N = \frac{H \times W}{P^2} \quad (2.1)$$

patches per frame.

Each patch $x_i \in R^{P^2 \cdot C}$ is then independently projected into a D -dimensional embedding space using a learnable linear projection denoted below:

$$z_i = x_i W + b, \quad W \in R^{(P^2 C) \times D} \quad (2.2)$$

producing a sequence of visual tokens for each frame. Over T number of frames, the encoder outputs:

$$\mathbf{V} \in R^{T \times N \times D} \quad (2.3)$$

This transforms visual information into token sequences, linking vision and language components. These visual tokens are not directly compatible with the language model.

The projection module solves this by mapping each visual token into the LLM's embedding space. This mapping is achieved by a learnable linear transformation:

$$x_i = z_i W_p, \quad W_p \in R^{D \times d} \quad (2.4)$$

where z_i is the visual token from the encoder and x_i is the aligned token passed to the language model.

After projection, the visual tokens are concatenated with the tokenised text prompt to pass a single input sequence to the LLM denoted as:

$$\mathbf{X} = [x_1, x_2, \dots, x_{TN}, t_1, t_2, \dots, t_M] \quad (2.5)$$

where x_i are visual tokens and t_j are text tokens. The LLM models the conditional probability:

$$P(\mathbf{y} \mid \mathbf{X}) = \prod_{t=1}^T P(y_t \mid y_{<t}, \mathbf{X}) \quad (2.6)$$

A key feature of this design is that visual and textual tokens share the same context window, allowing for joint reasoning. However, the main issue of Vid-LLMs, as aforementioned, is now clearly highlighted: The total number of tokens that can be processed within the context window is fixed.

2.3 The Long-Video Context Problem

The context window is a structure constraint in all transformer-based Vid-LLMs, arising from the shared token budget described in Section 2.2.5.

2.3.1 The Token Explosion

With a patch size of 16×16 pixels [9], a 224×224 frame produces 196 spatial tokens. Models such as LLaVA-Next [20] use dynamic high-resolution tiling, increasing the number of tokens per frame to 2,880. Assuming a sampling rate of 1 frame per second (1 fps), used in Video-ChatGPT [26], the total sequence length scales linearly with video duration.

$$L = T \times F \times V$$

where L is total number of tokens, T is video duration (seconds), F is sampling rate (fps), and V is the number of tokens per frame.

For a 45-minute video ($T = 2700$), LLaVA-Next requires:

$$L = 2700 \times 1 \times 2880 = 7,776,000 \text{ tokens}$$

Given LLaVA-Next’s 4,096-token context window [20], less than 0.06% of the entire video can be processed in a single forward pass. Figure 2.5 illustrates how quickly the context limit is reached, highlighting the fundamental mismatch between video duration and model capacity. For comparison, using a standard ViT configuration with $V = 196$ tokens per frame results in:

$$L = 2700 \times 196 = 529,200 \text{ tokens}$$

This allows only less than 1% of the video to be represented at once.

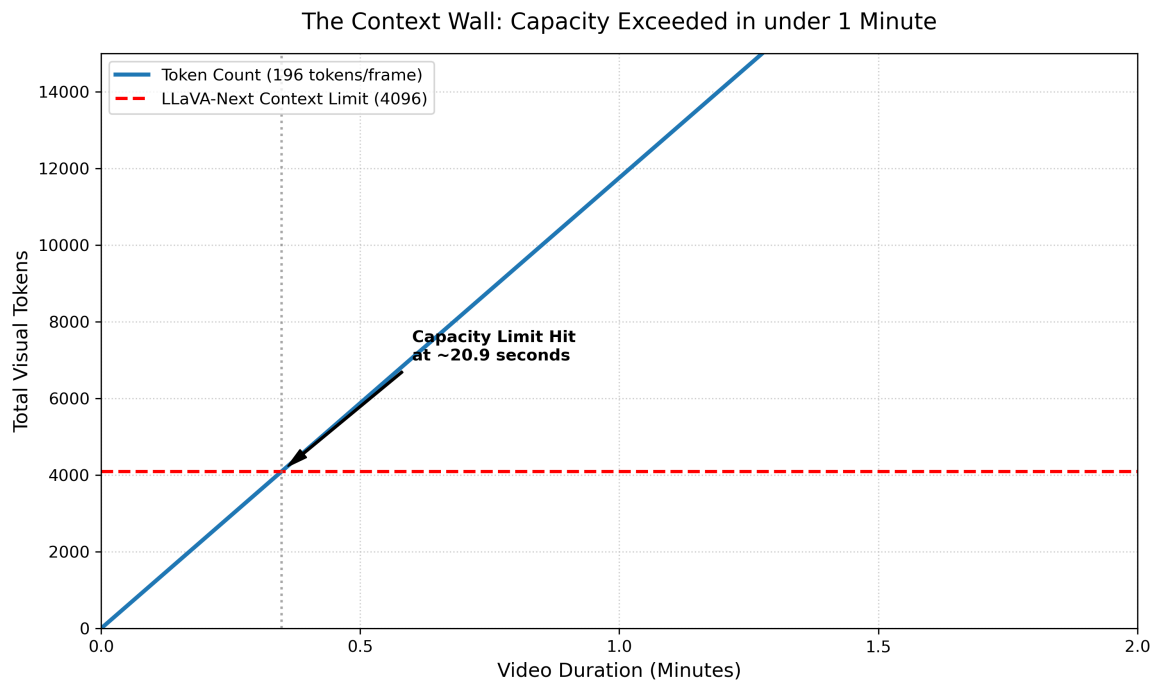


Figure 2.5: Linear growth of token length with video duration at a fixed 4,096 token context window. The limit is reached within seconds, demonstrating the infeasibility of processing long videos in a single pass.

Configuration	Tokens/frame	45-min total	% that fits in context
Standard ViT (16×16)	196	529,200	0.77%
LLaVA-Next tiling	2,880	7,776,000	0.053%
Context window limit	—	4,096	100%

Table 2.1: Token count comparison showing the mismatch between video length and LLaVA-Next’s context window.

The self-attention mechanism with transformer architecture requires each token to compare itself to every other token. The number of comparisons grows quadratically and can be denoted by $O(N^2)$ time and memory complexity [41]. Consequently, many Vid-LLMs aggressively downsample videos to fewer frames, discarding fine-grained details and missing moments that are needed to answer questions accurately. This mismatch between long videos and context windows motivates the approach to select only the most relevant frames.

2.3.2 Reducing the Token Explosion

This project examines a specific sampling method used by models such as Video-ChatGPT[26] and LLaVA-NeXT-Video[20] to reduce the token explosion. Uniform Sampling is a technique used to select a set of frames at equally spaced intervals across the video, which are then encoded, projected, and passed to the LLM. This technique is denoted by the equation below.

$$i_j = \left\lfloor \frac{j \times N}{k} \right\rfloor, \quad j = 0, 1, \dots, k - 1 \quad (2.7)$$

2.3.3 Limitations of Uniform Sampling

While uniform sampling is effective and simple to implement, it has two fundamental limitations. First, it is question-agnostic, so the same frames are selected regardless of the question, meaning that frames containing the specific visual evidence needed to answer a query may never be sampled. Second, the probability of missing the relevant segment requirement in long videos is high. For a 45-minute video with a 10-second evidence window and $k = 5$ frames sampled, the probability of missing the evidence is approximately 93.8%, as illustrated in Figure 2.6.

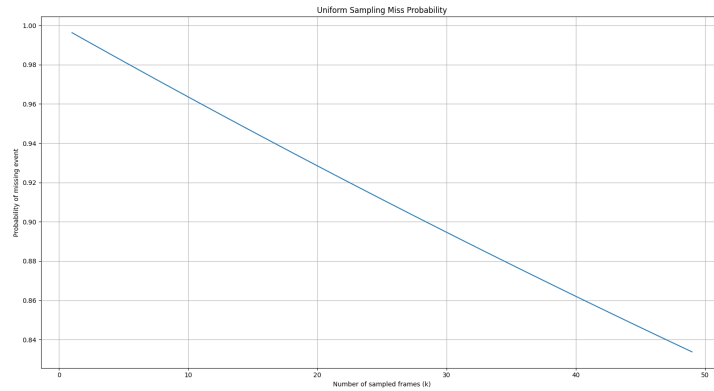


Figure 2.6: Uniform Sampling: Probability of missing relevant frame(s)

These limitations motivate a retrieval approach in which frame selection is dependent on the query.

2.4 Existing Approaches to the Context Problem

This section explores the current approaches to mitigate the context-bottleneck challenge within Vid-LLMs, structured into 3 key subsections:

1. **Question-Agnostic Compression:** focuses on reducing the large volume of data from videos.
2. **Question-Aware Selection:** focuses on choosing specific frames based on the semantics of the query
3. **Summary and Synthesis:** summarises the approaches explored and identifies the specific research gap addressed by this project.

2.4.1 Question-Agnostic Compression Approaches

Temporal Compression: MovieChat

The first key example is MovieChat [33], inspired by the Atkinson-Shiffrin [2] memory model of human cognition. The mechanism works by processing frames through a sliding window that separates visual information into Short-Term Memory (STM) and Long-Term Memory (LTM).

As shown in Figure 2.7, frames are converted into tokens; once the 18-frame STM capacity is met, tokens are compressed into a 256-token LTM buffer through a merging process that iteratively merges the most similar frame pairs. This architecture allows MovieChat to handle video sequences exceeding 10,000 frames, which was a scale previously unattainable for end-to-end transformers. In terms of efficiency, MovieChat manages a VRAM cost of approximately 21.3KB per frame compared to 227MB for standard Vid-LLMs, resulting in a 10,000× advantage [33].

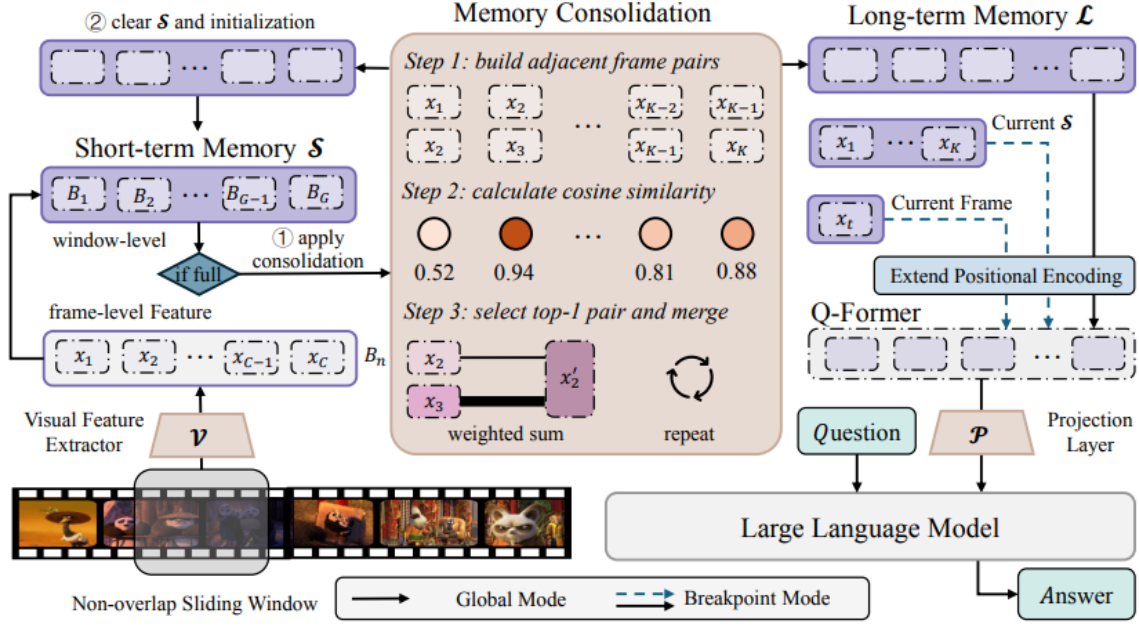


Figure 2.7: Illustration of MovieChat [33]

However, this technique introduces some fundamental flaws. To begin, the entire memory consolidation process is question-agnostic, meaning frames are merged based on visual similarity, with no consideration of the query. MovieChat provides only rough estimates of event durations in long videos, leading to a lack of precision in temporal details. Moreover, when two similar frames are merged into a single encoded token, any frame containing a precise moment or detail needed to answer a specific question may be permanently discarded. This effectively renders temporal compression unfit for question-answering tasks that require specific visual details, a challenge addressed by the retrieval-augmented pipeline.

Uniform Sampling with Pooling: Video-ChatGPT

Video-ChatGPT [26], a Vid-LLM that manages a long-form video by integrating a CLIP visual encoder with a Vicuna language decoder. To address the context bottleneck, it uses a dual-pooling strategy to reduce the number of visual tokens produced by CLIP:

- **Spatial Pooling:** Averages the patch tokens within each individual frame, resulting in a single descriptor per frame in the spatial dimension.
- **Temporal Pooling:** Averages the frame-level descriptors across the temporal dimension, resulting in a fixed-length representation that captures the overall meaning of a video.

As illustrated in Figure 2.8, these representations are concatenated and projected into the

LLM’s embedding space, allowing for an output. This efficiency allows the 7-billion-parameter model to require only 3 hours of training on 100,000 instruction-following pairs.

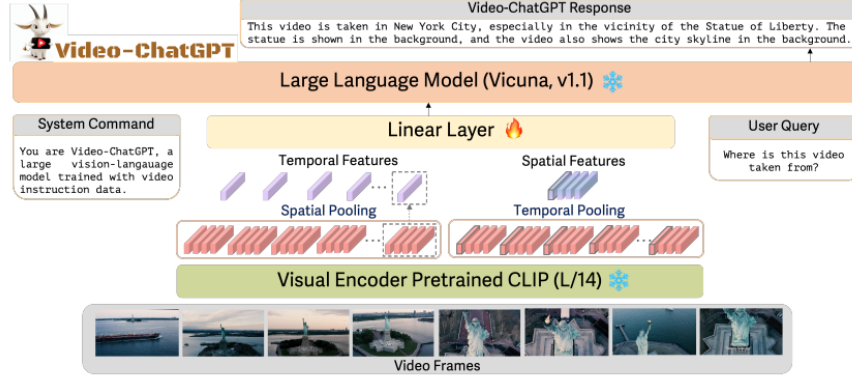


Figure 2.8: Video-ChatGPT Architecture [26]

While this technique efficiently captures both spatial and temporal aspects of a video, the fundamental limitation lies in the initial sampling decision. Maaz et al. [26] acknowledge that the model struggles to understand subtle temporal relationships in long videos exceeding two minutes. Furthermore, the model frequently fails to recognise the details of small objects, missing critical information. Hence, any fine-grained actions or visual cues between these sampled frames are completely lost before the encoder begins processing. This gap between sampled frames and the full video makes uniform sampling insufficient for long-form question answering and motivates a retrieval approach in which frame selection is based on the query.

Token-Level Compression: LLaVA-PruMerge

LLaVA-PruMerge[31] addresses the token limit by reducing the actual number of visual tokens produced. This differs from the previous inter-frame methods; PruMerge identifies and merges redundant spatial tokens (illustrated in Figure 2.9).

The model leverages attention scores from the CLIP encoder and analyses the attention weights between the CLS token and spatial patch tokens to identify redundant patches.

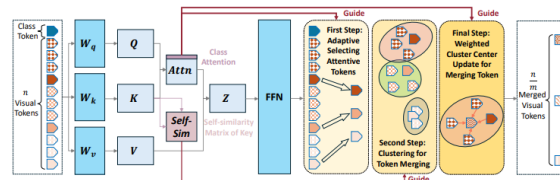


Figure 2.9: LLaVA-PruMerge architecture: Attention-based token selection and k-NN merging [31].

PruMerge uses k-NN (k-nearest neighbours) clustering to merge these patches while preserving global spatial information.

While this model applies techniques at a different level from the proposed retrieval-based approach, it compresses information within the encoded representation rather than selecting frames before encoding. Token selection is performed entirely by the encoder’s internal attention scores with no reference to the user query; a token deemed unimportant during encoding may be the visual evidence needed to answer a given query.

2.4.2 Question-Aware Selection Approaches

Learned Keyframe Selection: SeViLA

The SeViLA (Self-Chained Video Localization Answering)[48] model demonstrates a shift towards question-aware selection. It uses the user query to actively reduce the video down to only the most relevant moments. The model handles both temporal key-frame localisation and question answering on videos using two modules: a Localizer and an Answerer through parameter fine-tuning of BLIP-2 [21] (see Figure 2.10).

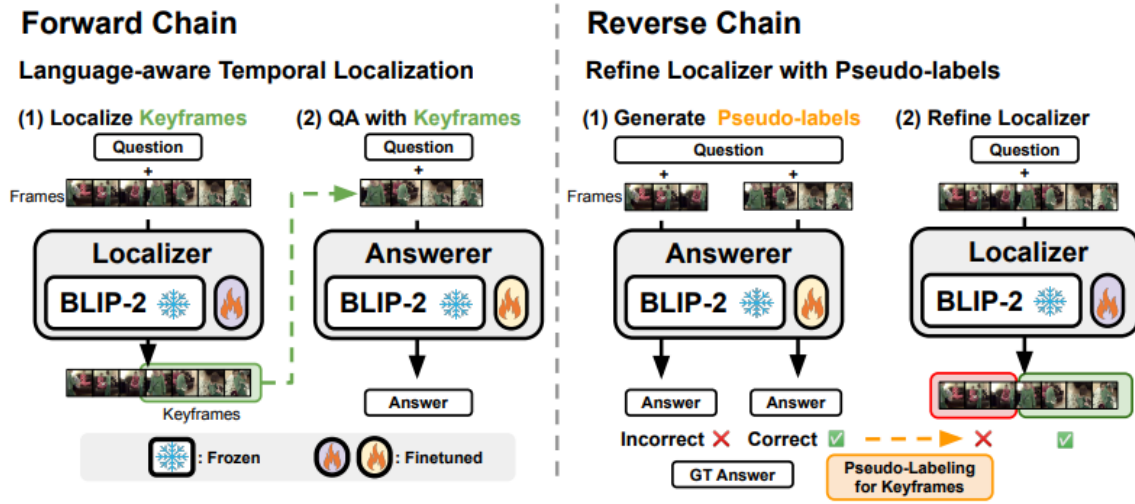


Figure 2.10: The SeViLA framework: Localization and Answering modules sharing BLIP-2 parameters [48].

The mechanism works as follows:

- Localizer evaluates 32 sampled frames against a prompt containing the query.
- Each frame is assigned a similarity score, and the top $K = 4$ frames with the highest scores are selected as key-frames.
- These key-frames are passed to the Answerer module

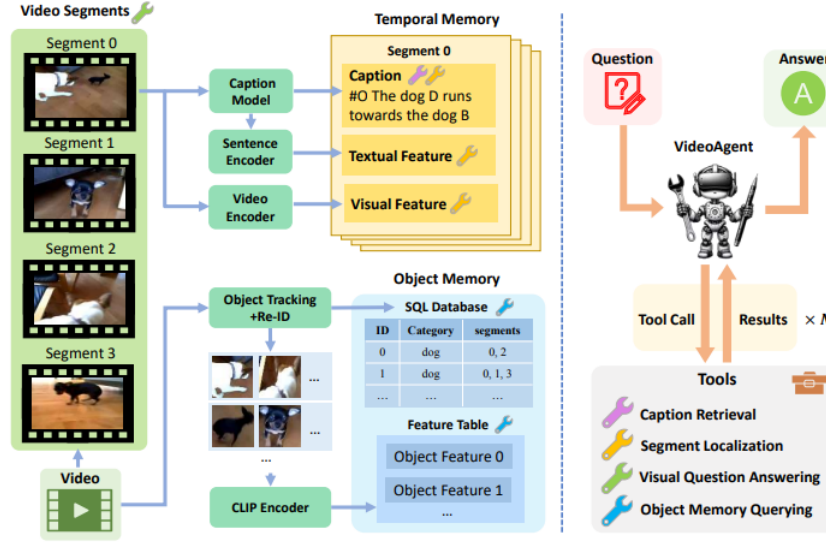


Figure 2.11: VideoAgent Architecture: LLM is prompted to solve a task by using tools that work with the Temporal and Object Memory

Through this mechanism, 87.5% of all visual data is discarded before reasoning begins, allowing SeViLA to achieve 63.3% accuracy on the NExT-QA benchmark [48] under zero-shot conditions.

The SeViLA model demonstrates that question-aware selection outperforms uniform sampling for long-video question answering. However, SeViLA relies on a complex training process, involving pre-training on the QVHighlights dataset, which contains 10,148 human-annotated videos [48], followed by fine-tuning of both modules. This requires labelled data and high computational power to align the Localizer. This project aims to explore a training-free approach that uses embeddings to achieve efficient retrieval without requiring fine-tuning or task-specific supervision.

Agent-Based Iterative Retrieval: VideoAgent

VideoAgent[10] is a training-free model that uses an LLM agent for frame retrieval. VideoAgent creates a temporal memory that stores text captions for each two-second segment from the video, and an object memory that tracks object occurrences across the video in a structured SQL schema (see Figure 2.11).

Given a query, a GPT-4 agent invokes tools such as a CLIP-based segment localisation [30], VQA (Video Question Answering), and caption retrieval, looping until it has gathered sufficient evidence to produce an answer.

While this training-free approach seems cost-effective, each query requires multiple sequential LLM calls via the GPT-4 engine, making large-scale evaluation across hundreds of videos both costly and rate-limited. The single-pass architecture of CLIP-based retrieval avoids this cost

entirely and is what this project will aim to retain and use.

2.4.3 Summary and Synthesis

Table 2.2 summarises the key properties of each existing approach against the scope of this project.

Approach	Operates at	Question-aware?	Training required?
MovieChat	Token compression	No	Yes
Video-ChatGPT	Frame selection + pooling	No	Yes
LLaVA-PruMerge	Token compression	No	No
SeViLA	Frame selection	Yes	Yes
VideoAgent	Frame selection (iterative)	Yes	No
This project	Frame selection	Yes	No

Table 2.2: Comparison of existing approaches to the long-video context problem across the scope of this project

Question-agnostic methods such as MovieChat and Video-ChatGPT achieve scalability through data compression and uniform sampling, but both discard visual data before considering the query. This is a flaw in question answering, as the most relevant frames may be discarded before reasoning begins. LLaVA-PruMerge operates at a finer granularity by compressing tokens within an encoded frame, but shares the same limitation. Token selection operates entirely on the encoder’s internal attention scores, with no awareness of the query.

Question-aware approaches address these limitations by using the query to improve question answering. SeViLA shows that question-aware frame selection outperforms uniform sampling, achieving 64.9% on the NExT-QA dataset. However, this performance requires pre-training on QVHighlights and downstream fine-tuning, increasing dependence on labelled data. VideoAgent does not require training as it utilises an LLM agent to iteratively retrieve visual evidence. However, each query requires multiple sequential calls to GPT-4, making large-scale evaluation unsustainable.

Therefore, across all five existing approaches, no single method satisfied all three requirements: question-awareness, training-free, and single-pass retrieval (see Table 2.2). The proposed pipeline addresses this gap directly: CLIP provides zero-shot cross-modal retrieval without fine-tuning, FAISS enables single-pass nearest-neighbour search, and the training-free design requires no task-specific supervision.

2.5 Tools for Visual Retrieval

The research gap identified in Section 2.4 calls for a retrieval mechanism that is question-aware, training-free, and single-pass. This section justifies selecting CLIP [30] as the visual encoder for that retrieval mechanism.

2.5.1 Visual-Semantic Alignment through Contrastive Learning

A core requirement for the retrieval stage is a shared embedding space: a vector space in which a text query and a relevant frame are geometrically close, while the query and an irrelevant frame are distant. This is known as visual-semantic alignment and must be learned from paired data.

Contrastive learning [18] is the dominant training technique for achieving visual-semantic alignment. Given a batch of N input pairs, the model is trained to maximise the similarity score between matched pairs (positive pairs) and minimise the similarity score between mismatched pairs (negative pairs). This objective is denoted as:

$$\mathcal{L} = -\log \frac{\exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_j)/\tau)}{\sum_{k=1}^{2N} \mathbf{1}_{[k \neq i]} \exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_k)/\tau)}$$

where z_i and z_j are the projected representations of a matched input pair, τ is a temperature hyper-parameter controlling the sharpness of the distribution, and the denominator sums over all $2N$ samples in the batch.

Figure 2.12 illustrates this alignment process of learning similarity between images of instruments and their respective sounds. The result of this training is an embedding space in which semantic similarity is encoded, making it suitable for nearest-neighbour retrieval without any task-specific fine-tuning. This property makes contrastively trained encoders well-suited to the zero-shot retrieval aspect of the pipeline.

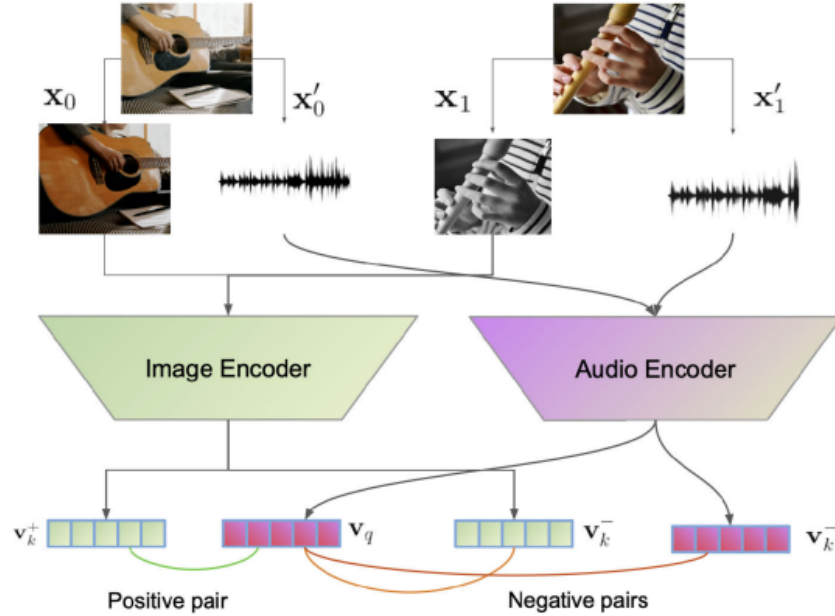


Figure 2.12: Illustration of learning similarity between multiple modalities [18]

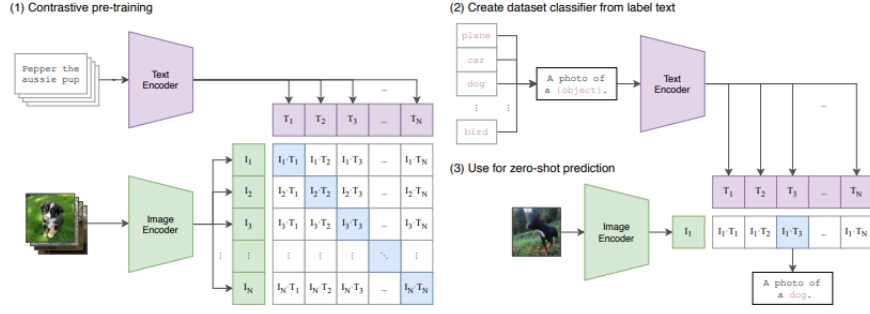


Figure 2.13: CLIP Architecture [30]

2.5.2 The CLIP Architecture

CLIP (Contrastive Language-Image Pre-Training)[30] is a vision-language model developed by Radford et al and applies contrastive learning at scale across two modalities: image and text. The model is trained on WIT [30], a dataset containing 400 million image-text pairs collected from the internet, enabling it to learn a rich, shared embedding space across a broad range of visual concepts.

The architecture consists of two encoders, an image encoder and a text encoder that project inputs into a D-dimensional vector space. The model has to classify $N \times N$ (image, text) pairs and predict which pairs are a 'real' match. This is achieved by maximising the similarity between correct pairs (positive pairs) and minimising the similarity between $N^2 - N$ mismatched pairs (negative pairs). CLIP can perform zero-shot retrieval without any task-specific fine-tuning because of this resulting embedding space. An illustration is shown in Figure 2.13

$$\mathcal{L}_{\text{CLIP}} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(\cos(\mathbf{v}_i, \mathbf{t}_i)/\tau)}{\sum_{j=1}^N \exp(\cos(\mathbf{v}_i, \mathbf{t}_j)/\tau)}$$

CLIP as the Vision Encoder

CLIP was ultimately chosen as the encoder over alternative options for three reasons. First, its zero-shot capability was a requirement of the training-free constraint identified in the research gap. Second, CLIP's ViT-B/32 variant produces compact 512-dimensional embeddings, balancing retrieval quality with the computational cost of encoding an entire video. Third, and most importantly, previous work has directly validated the use of CLIP as a visual encoder. For instance, CLIP4Clip demonstrates that frame-level embeddings are sufficient for high-quality retrieval, and even VideoAgent [10], one of the strongest baselines reviewed in Section 2.4, uses CLIP. These reasons validate the choice of selecting CLIP.

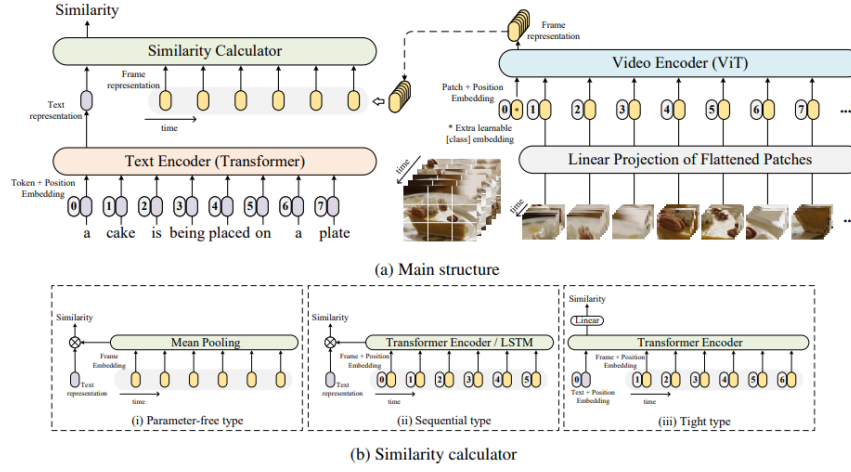


Figure 2.14: CLIP4Clip Architecture

2.5.3 Extending CLIP to Video: CLIP4Clip

The CLIP architecture is better suited to static images and cannot be used directly for videos without adaptation. The CLIP4Clip[25] model explores how the architecture can be used for videos by treating each video as a sequence of independently encoded frames using the same ViT-B/32 image encoder from the original CLIP[30] architecture.

The encoder has two modes: a 2D mode that treats each frame independently and converts each frame to produce a 512-dimensional embedding vector, and a 3D mode that captures temporal features across patches from multiple frames simultaneously, as illustrated in Figure 2.14. The authors found that the 2D mode outperformed the 3D mode during initialisation, demonstrating that frame-level independent encoding is sufficient for high-quality retrieval. This key finding justifies the use of CLIP for this project.

Known Limitations of CLIP for Video Retrieval

CLIP encodes each frame independently and produces no representation of motion, temporal order, or causal relationships between frames [30]. Each frame embedding only captures the static visual content, with no awareness of what happened before or after.

For object recognition or attribute recognition tasks, static frame embeddings are sufficient as the visual evidence exists within the single frame, and CLIP can locate it through cross-modal similarity. However, for temporal reasoning tasks, such as identifying actions across frames, a static encoder cannot capture the evidence needed to answer the question. This reduces retrieval precision for queries about dynamic events, leading to weaker outputs for temporal reasoning tasks.

2.6 Vector Search and Similarity Metrics

Frame embeddings and the query embedding produced by CLIP need to have a similarity score calculated to select the most relevant frame. This requires a similarity metric that captures semantic proximity in the embedding space and an indexing structure that enables efficient search.

2.6.1 Introducing FAISS

FAISS (Facebook AI Similarity Search) [14] is a library used for efficient similarity search designed over high-dimensional vectors. For this project IndexFlatL2 is used, which applies exact nearest-neighbour search using Euclidean distance over all stored embeddings. However, CLIP frame embeddings are L2-normalised before being stored, essentially meaning that Euclidean distance and cosine similarity measure vectors identically on normalised vectors. Take two unit vectors $\mathbf{a}$ and $\mathbf{b}$:

$$\|\mathbf{a} - \mathbf{b}\|^2 = 2 - 2 \cos(\mathbf{a}, \mathbf{b}) \quad (2.8)$$

Since 2 is a constant, minimising $\|\mathbf{a} - \mathbf{b}\|^2$ is equivalent to maximising $\cos(\mathbf{a}, \mathbf{b})$. IndexFlatL2 therefore functions as an exact cosine similarity search on L2-normalised embeddings.

For search methods such as the Inverted File Index (IVF), the embedding space is partitioned into clusters, restricting the search to the nearest clusters. Since the largest FAISS index in this project contains at most 3,600 vectors (one per second for a 60-minute video), an exact search is completed in milliseconds and introduces no meaningful latency. The additional complexity of approximate indexing is therefore unnecessary; exact search is both simpler and more accurate at this scale.

2.7 Downstream LLMs for Reasoning

2.7.1 Model Selection

The proposed pipeline uses LLaVA-Next [20] for the downstream question-answering component. This model is a vision-language model that accepts an image and a text prompt as input and generates a text response. The following subsection will explore the architecture and properties of LLaVA-Next that make it suitable for this task, and recap the context window constraint that influenced key design decisions in the pipeline.

First, LLaVA-Next is openly available under a permissive research license through HuggingFace. Proprietary models such as GPT-4V are accessed via an API, introducing costs and rate limits that make large-scale benchmark evaluation impractical for this project.

The chosen 13B-parameter version of LLaVA-Next operates within the 40GB of VRAM available on the A100 GPU. Larger models such as LLaVA-NeXT-Video 34B would require

more VRAM from the available A100 GPU. Furthermore, LLaVA-Next is among the models evaluated in the Video-MME benchmark [12], providing published baseline results against which the pipeline’s results can be directly compared.

2.7.2 LLaVa-Next Architecture

LLaVA-Next follows the standard Vid-LLM architecture discussed in Section 2.2.5: a ViT visual encoder projects patches into embeddings, a projection layer aligns with the LLM’s token space, and the concatenated visual and text token sequence is passed to a Vicuna-based language model [20] for response generation. The prompt is structured as follows: USER `[image]` `[question]` ASSISTANT: and the model provides the answer. For this particular use case, the question and options are provided as text, better suiting the Vicuna format.

2.7.3 Context Window Constraint and Frame Tiling

LLaVA-Next has a maximum context length of 4,096 tokens [20]. The visual encoder produces an image token sequence, with each token containing approximately 576 tokens, assuming spatial pooling is applied. With $k = 5$ retrieved frames, passing each frame as a token would cost roughly 2,880 visual tokens, leaving only 1,216 tokens for the questions, options, and generated answer. This is simply insufficient, especially for multiple questions.

To address this issue, the pipeline implements a frame-tiling strategy: retrieved frames are resized to 336×336 pixels and stitched into a single grid image, which is then passed as a single image input. This approach is inspired by the multi-frame tiling used in LLaVA-NeXT-Video [20], ensuring that the total token count remains within the context limit while allowing the model to reason over all retrieved frames. The pipeline evaluates both the top-ranked retrieved frame and the top-5 retrieved frames using tiling to analyse the contribution of multi-frame context.

2.8 Summary and Research Gap

This chapter has identified and developed the technical case for the proposed pipeline based on the surveyed literature. Three main conclusions can be made:

1. The context bottleneck has highlighted that frame selection should be based on the question rather than uniform sampling, given the architecture of Vid-LLMs.
2. Existing selection strategies explored through prior approaches present a fundamental trade-off between efficiency and query relevance. Question-agnostic methods aim to discard a large amount of visual content before the query is considered. However, this is a flaw for tasks requiring specific visual evidence. Question-aware methods resolve this by selecting frames based on the query. However, this requires supervised

pre-training or sequential LLM calls as seen in prior work. Ultimately, both approaches are unsuitable for training-free, single-pass, and large-scale evaluation.

3. As shown in Table 2.2, there is no existing method that is simultaneously question-aware, training-free, and single-pass. The core hypothesis of this project is that this combination will achieve higher question-answering accuracy than a uniform-sampling baseline. Chapter 3 discusses how the pipeline is designed to test this hypothesis, and Chapter 5 presents the empirical results.

Chapter 3

Approach

3.1 Research Questions and Goals

The primary research question motivating this project, established in Section 1.2, is re-stated below:

Does semantic frame retrieval improve multiple-choice question-answering accuracy on long-form videos compared to uniform sampling, and if so, under which conditions (question types, video domain, and video duration)?

To address this, three sub-questions are derived, each targeting a distinct aspect of retrieval performance.

1. Does CLIP-based semantic retrieval outperform the uniform sampling baseline overall on the Video-MME benchmark?
2. Does the performance gap between semantic retrieval and uniform sampling increase as video duration grows, consistent with the prediction in Section 2.3.3?
3. Are there specific video domains or question types where semantic retrieval outperforms or underperforms uniform sampling?

Sub-question 1 establishes whether the proposed pipeline creates a meaningful advantage over the baseline. Sub-question 2 tests the core hypothesis that the benefit of question-aware retrieval scales with video duration. Sub-question 3 identifies the specific conditions under which semantic retrieval is most and least effective.

3.2 Proposed Approach

Chapter 2, Section 2.4 established the research gap, which highlights that no existing method simultaneously satisfies all three requirements for large-scale video question-answering: question-awareness,

training-free, and single-pass retrieval. The proposed CLIP-based pipeline is designed to directly address this gap.

Given a video and a natural language query, the pipeline retrieves the top- k most semantically relevant frames and passes them to a vision-language model for downstream multiple-choice question answering. This contrasts with the standard approach of using uniform sampling to select frames at fixed intervals regardless of the query.

3.2.1 Justification for this Approach

Table 2.2 demonstrates that existing methods fail to satisfy all three requirements simultaneously. Question-agnostic methods such as MovieChat [33], Video-ChatGPT [26], and LLaVA-PruMerge [31], all achieve scalability by removing visual data before the query is considered, making it impossible to guarantee that frames containing query-relevant visual evidence are retrieved. Question-aware methods such as SeViLA [48] and VideoAgent [10] address this issue through the use of question-aware techniques, but at the cost of reliance on labelled data and multiple sequential LLM calls.

The proposed pipeline addresses all three requirements. CLIP [30] allows for a zero-shot shared embedding space that allows text queries and video frames to be compared directly without fine-tuning. FAISS [14] provides nearest-neighbour retrieval in a single pass. LLaVA-Next [20] allows for open, reproducible downstream question-answering reasoning within a fixed hardware budget. No component in the pipeline requires fine-tuning, and it executes in a single forward pass per query.

3.2.2 Pipeline Architecture

The pipeline is divided into three overarching stages as illustrated in Figure 3.1.

Stage 1 Preprocessing and Indexing: The video is downloaded from the specified YouTube URL, and frames are extracted at 1 FPS (1 frame per second). Each frame is then encoded into a 512-dimensional embedding using CLIP ViT-B/32 and stored in a FAISS index. This occurs once per video and is cached after completion.

Stage 2 Retrieval: Given a user query, the two retrieval strategies are applied. The uniform sampling baseline selects $k = 5$ frames at evenly spaced intervals, regardless of the query. The semantic retrieval strategy encodes the query as a CLIP text embedding and retrieves the top $k = 5$ most semantically similar frames using FAISS, alongside temporal windowing.

Stage 3 Multi-modal Reasoning: For the tiled mode, the top- k retrieved frames are arranged into a grid image and passed as a single visual input to LLaVA-Next 13B alongside the question and answer options. For the single mode, only the highest-ranked frame is passed onto the LLM.

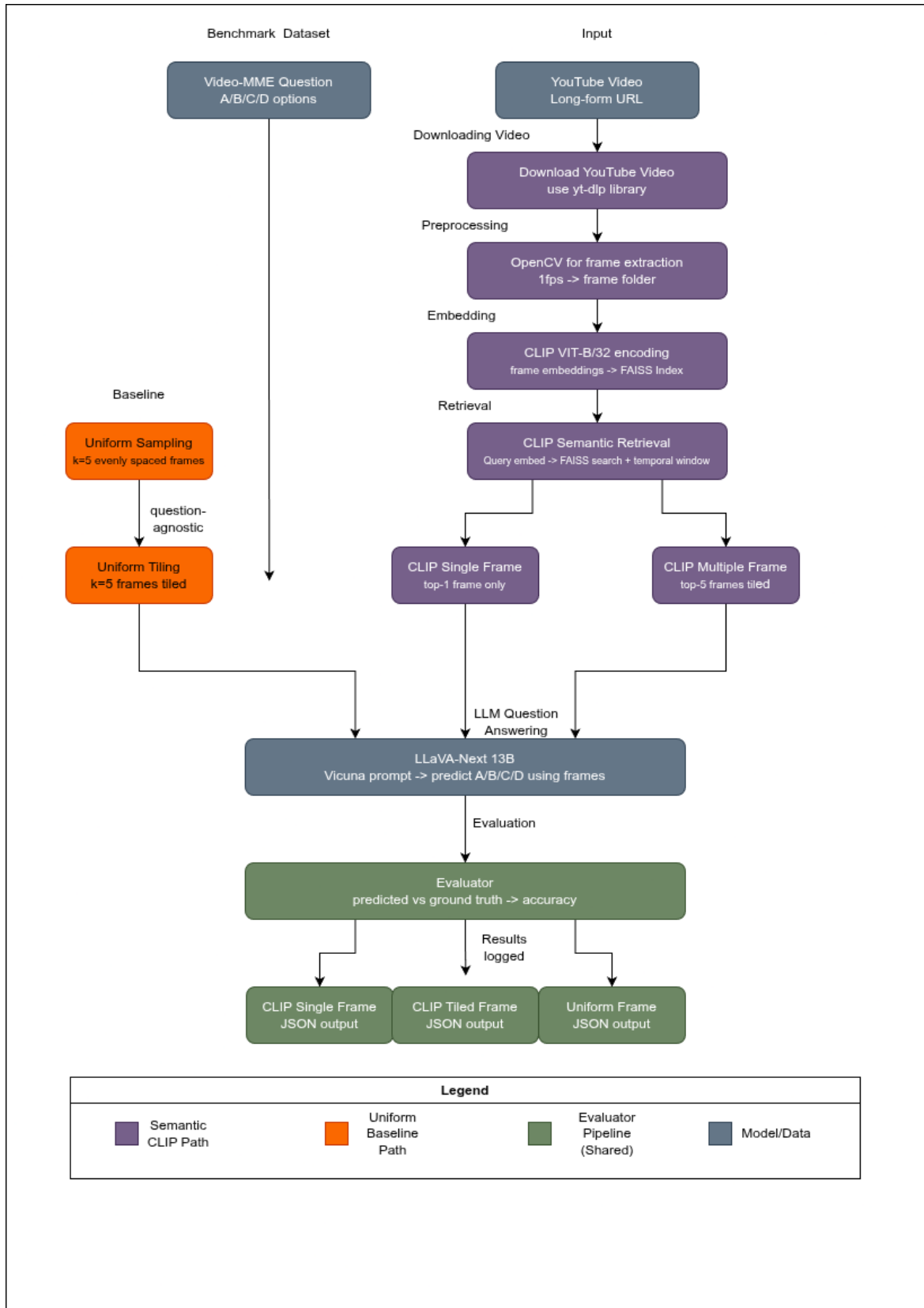


Figure 3.1: The pipeline architecture, split into three stages: Pre-processing and Indexing, Retrieval, and Multi-modal Reasoning.

3.2.3 Design Principles

There are three core properties of the pipeline influenced by the research gap identified in Section 2.4: training-free, query-aware, and single-pass. Each property is necessary for the pipeline to be both a valid contribution towards the field and a viable evaluation system.

Training Free

The retrieval pipeline does not require any fine-tuning or task-specific supervision. As discussed in the Section 2.4, approaches such as SeViLA [48] require expensive pre-training and fine-tuning on annotated datasets, creating dependence on labelled data. By relying on pre-trained CLIP embeddings for retrieval and a pre-trained LLaVA-Next model for reasoning, the pipeline can be applied to any video in any domain without modification.

Query-Based Selection

Each selected frame is based on the specific question asked. As highlighted in the review of MovieChat [33] and Video-ChatGPT [26], question-agnostic techniques discard visual content before the query is ever used. This risks permanently losing frames that may contain the visual evidence needed to answer the question. The proposed retrieval pipeline encodes the query as a text embedding and semantically retrieves the most relevant frames, ensuring that selection depends on the query rather than on temporal position.

Single-pass

The pipeline retrieves frames in a single forward pass through the FAISS index. As identified in the literature review, VideoAgent [10] requires multiple sequential LLM calls per query, making large-scale evaluation inefficient. The single-pass method avoids this challenge entirely and ensures the pipeline is tractable for large-scale benchmark evaluation.

3.2.4 Module Overview

The six Python modules and their respective functions are summarised in Table 3.1.

Module	File	Role
Loader	<code>loader.py</code>	Loads Video-MME from HuggingFace, filters by video ID, returns questions and ground truth answers
Preprocessor	<code>preprocessor.py</code>	Downloads video via yt-dlp. Extracts frames at 1 FPS using OpenCV
Indexer	<code>indexer.py</code>	Encodes frames using CLIP ViT-B/32 using SentenceTransformer. Stores embeddings in a FAISS IndexFlatL2 index
Retriever	<code>retriever.py</code>	Implements both semantic retrieval with temporal windowing and uniform sampling baseline
Responder	<code>responder.py</code>	Runs the LLaVA-Next 13B model using the Vicuna prompt format
Evaluator	<code>evaluator.py</code>	Executes both retrieval methods per question, compares predictions to ground truth, saves JSON results.

Table 3.1: The six modules of the RAG pipeline and their roles.

3.3 Algorithmic Design

This section explains the pipeline at the algorithmic level, without including specific software or hardware. Each stage is described in terms of inputs, core operations, design justification, and outputs. The use of tools and software is described in Section 3.4.

3.3.1 Stage 1: Frame Extraction

Take a video of duration T seconds with a frame rate of r frames per second. A frame is saved every $\lfloor r \rfloor$ raw frames to a frames directory, where $\lfloor \cdot \rfloor$ denotes rounding to the nearest integer. This handles non-integer frame rates such as 29.97 fps correctly. The result is a sequence of $N \approx T$ frames:

$$\mathcal{F} = \{f_0, f_1, \dots, f_{N-1}\} \quad (3.1)$$

where the index of each frame is the timestamp of that frame in seconds.

The final output is a set of N frames $\mathcal{F}$, saved to disk as **frame_XXXX.jpg**, where the four-digit index corresponds to the frame’s timestamp in seconds.

Two considerations motivated this choice. First, using a fixed extraction rate ensures a standardised time resolution throughout the analysis of all videos in the benchmarks, even though videos may support a wide range of frame rates and durations. This ensures that comparisons between retrieval strategies are not affected by differences in frame density.

Second, an extraction rate of 1 FPS substantially reduces the number of frames that must be encoded and indexed: a 30-minute video produces 1,800 frames, which is computationally tractable for CLIP encoding and FAISS indexing while maintaining enough frame coverage for efficient retrieval. However, this introduces a trade-off, as reducing the number of frames may degrade performance for videos with action-heavy content.

3.3.2 Stage 2: Embedding and Indexing

Each frame $f_i \in \mathcal{F}$ is passed through a visual encoder ϕ_v to produce a 512-dimensional embedding:

$$\mathbf{e}_i = \phi_v(f_i), \quad \mathbf{e}_i \in R^{512} \quad (3.2)$$

Each embedding is then L2-normalised, creating a set of normalised embeddings. The full set of normalised embeddings is stacked into a matrix $E \in R^{N \times 512}$ and added into a nearest-neighbour index $\mathcal{I}$. As established in Section 2.6.1, L2-normalisation ensures that Euclidean distance over the index $\mathcal{I}$ is mathematically equivalent to cosine similarity.

$$\hat{\mathbf{e}}_i = \frac{\mathbf{e}_i}{\|\mathbf{e}_i\|_2} \quad (3.3)$$

The encoder ϕ_v needs to produce frame embeddings that are comparable to the text query embeddings, enabling cross-modal similarity search without fine-tuning. As justified in Section 2.5, CLIP satisfies this requirement through its contrastive pre-training on 400 million image-text pairs and is the reason why it was chosen. Frames are encoded independently rather than jointly, which CLIP4Clip [25] demonstrated is sufficient for high-quality video retrieval.

The index $\mathcal{I}$ is constructed once per video and saved to disk. Before building an index, the pipeline checks whether an index already exists for a specific video ID and skips encoding if it does. This is important given the Video-MME benchmark structure, where each video contains three questions. Caching the index reduces encoding cost from three passes to one, as an index for a video only has to be built once.

3.3.3 Stage 3: Retrieval

The retrieval stage is the central contribution of this project, and two strategies are implemented and compared. Both operate on the same index $\mathcal{I}$ and use the same frame budget $k = 5$, ensuring that the retrieval strategy is the only variable that changes.

Strategy A: Semantic Retrieval with Temporal Windowing

A query string Q and a list of four answer option strings $O = \{A, B, C, D\}$ from the Video-MME benchmark [12], and the index $\mathcal{I}$ are provided as input. The query and all

four options are concatenated into a single query string:

$$q = Q \parallel A \parallel B \parallel C \parallel D \quad (3.4)$$

where $\parallel$ denotes string concatenation. This string is passed through CLIP’s text encoder ϕ_t and L2-normalised:

$$\hat{\mathbf{q}} = \frac{\phi_t(q)}{\|\phi_t(q)\|_2}, \quad \hat{\mathbf{q}} \in R^{512} \quad (3.5)$$

Encoding the question alone produces an embedding that reflects only the question’s semantic content. The reason for including all four answer options is that the query embedding is enhanced by the visual concepts referenced across all choices, increasing the likelihood that retrieved frames contain the visual evidence needed to choose the correct option.

The normalised query embedding $\hat{\mathbf{q}}$ is passed to the index $\mathcal{I}$, returning the top- $4k$ most similar frames as a candidate set:

$$\mathcal{C} = \text{topK}_{4k}(\mathcal{I}, \hat{\mathbf{q}}) \quad (3.6)$$

A known failure mode of nearest-neighbour retrieval over video frames is temporal clustering, where the top-ranked frames are clustered around a single moment in a video, returning near-identical frames rather than a diverse set. To address this, a minimum temporal gap δ is computed from the video length:

$$\delta = \max\left(30, \left\lfloor \frac{N}{2k} \right\rfloor\right) \quad (\text{seconds}) \quad (3.7)$$

where N is the total number of extracted frames and k is the target frame count.

This scales the gap to the video’s duration: for a 30-minute video ($N = 1800$, $k = 5$), $\delta = \max(30, 180) = 180$ seconds; for a short video ($N = 120$), $\delta = \max(30, 12) = 30$ seconds. The floor of 30 seconds ensures that a minimum diversity constraint is applied even for very short videos.

A greedy filter is then applied to $\mathcal{C}$ using Algorithm 1.

where $\text{ts}(f)$ is the timestamp in seconds of frame f , extracted from its filename. The fallback at lines 11–18 handles edge cases where the diversity constraint is too strict relative to the video’s length, using the candidate set to fill the remaining slots, ensuring exactly k frames are always returned.

This outputs a set $\mathcal{S}$ of $k = 5$ temporally diverse frames, ranked by relevance to the query.

Algorithm 1 Adaptive Temporal Windowing

Require: Candidate set $\mathcal{C}$ (top- $4k$ frames, ranked by cosine similarity to query), target count k , adaptive gap δ

Ensure: Selected set $\mathcal{S}$ of up to k temporally diverse frames

```

1:  $\mathcal{S} \leftarrow \emptyset$ 
2:  $T_{\mathcal{S}} \leftarrow \emptyset$  ▷ timestamps of selected frames
3: for each frame  $f \in \mathcal{C}$  in order of decreasing similarity do
4:   if  $|\text{ts}(f) - t| \geq \delta$  for all  $t \in T_{\mathcal{S}}$  then
5:      $\mathcal{S} \leftarrow \mathcal{S} \cup \{f\}$ 
6:      $T_{\mathcal{S}} \leftarrow T_{\mathcal{S}} \cup \{\text{ts}(f)\}$ 
7:   end if
8:   if  $|\mathcal{S}| = k$  then
9:     break
10:  end if
11: end for
12: if  $|\mathcal{S}| < k$  then ▷ Fallback: relax gap constraint
13:   for each frame  $f \in \mathcal{C}$  do
14:     if  $f \notin \mathcal{S}$  then
15:        $\mathcal{S} \leftarrow \mathcal{S} \cup \{f\}$ 
16:     end if
17:     if  $|\mathcal{S}| = k$  then
18:       break
19:     end if
20:   end for
21: end if
22: return  $\mathcal{S}$ 

```

Strategy B: Uniform Sampling Baseline

The uniform baseline contains the ordered frame set $\mathcal{F}$ of size N , and the frame budget $k = 5$. Frames are selected at k evenly spaced positions across the full index range $[0, N - 1]$:

$$i_j = \text{round}\left(\frac{j \times (N - 1)}{k - 1}\right), \quad j = 0, 1, \dots, k - 1 \quad (3.8)$$

This outputs a set of $k = 5$ frames at evenly spaced temporal positions, identical across all queries for a given video.

3.3.4 Stage 4: Multi-modal Reasoning

The pipeline supports two modes alongside the uniform sampling baseline to present frames to the model, corresponding to the two experimental conditions evaluated:

Single mode (clip_single): Only the highest-ranked frame in set $\mathcal{S}$ of $k = 5$ (the frame most similar to the query) is passed to the model as a single image. This mode serves as an ablation condition to isolate the contribution of the tiling strategy by comparing it against the single best-matched frame.

Tiled mode (clip_tiled and uniform): All k frames are sorted into chronological order by their timestamps, then resized to 336×336 pixels each and concatenated into a 3-column grid image. For $k = 5$, this produces a 3×2 grid (1008×672 pixels) with one empty black cell in the bottom-right position. The grid image is passed to the model as a single image input, keeping the total visual token count within the context limit while making all retrieved frames visible to the model.

The composed image and question text are passed to LLaVA-Next [20] using the Vicuna prompt format. The prompt includes a parameter for image description to direct the model to the type of visual input it is receiving:

- **Single mode:** *"This is a single frame extracted from a video at the most semantically relevant moment for the question below."*
- **Tiled mode:** *"This image contains [k] frames extracted from a video, arranged in chronological order from left to right and top to bottom. Each cell represents a different moment in the video. Use all frames together as visual evidence to answer the question below."*

The full prompt structure is:

```
USER: <image>
[image description]
Select the best answer to the following multiple-choice question based
on the video. Respond with only the letter (A, B, C, or D) of the correct
```

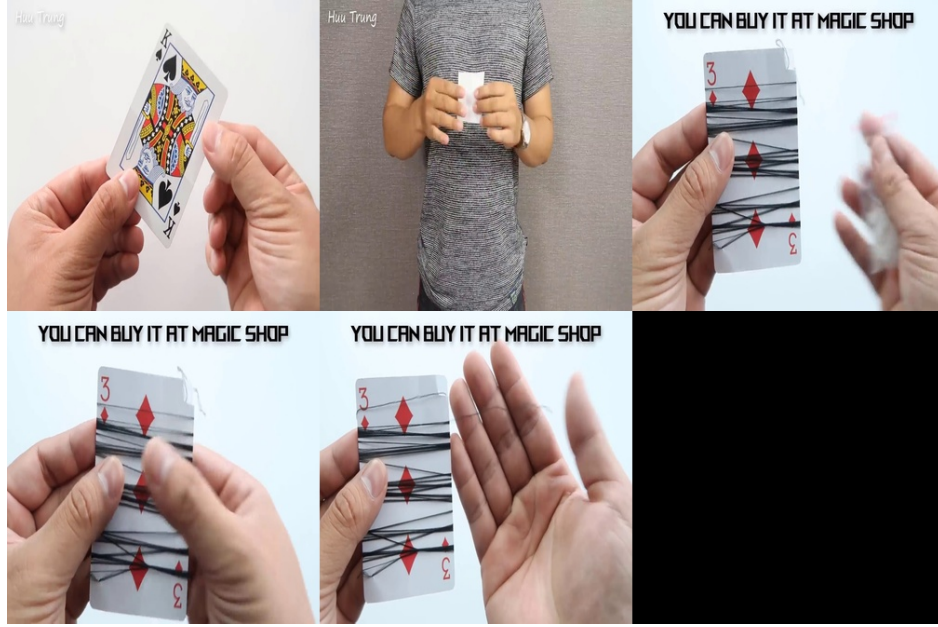


Figure 3.2: Tiled frame composition for $k = 5$ frames. Frames are sorted chronologically and arranged into a 3×2 grid at 336×336 pixels per cell (1008×672 pixels total). The bottom-right cell is black-filled. The composite is passed to LLaVA-Next as a single image input.

option.

Question: [question text]

A. [option A]

B. [option B]

C. [option C]

D. [option D]

The best answer is:

ASSISTANT:

This prompt structure follows the standard established in the Video-MME benchmark paper [12], ensuring valid comparisons can be made with published baselines.

The model generates up to 10 new tokens under greedy decoding (`do_sample=False`, `temperature=0`), ensuring fully deterministic output. The decoded string is then scanned for the first occurrence of a valid answer letter (A, B, C, or D) regardless of case-sensitivity. If no valid letter is found, the prediction is recorded as X and counted as incorrect. Greedy decoding is used throughout to ensure that all results are reproducible.

The LLaVA-Next model is loaded once into GPU memory and shared across both the single and tiled responder instances. This avoids loading the 13-billion-parameter model twice, which would exceed the A100 GPU's 40GB VRAM capacity. The final output is a predicted answer letter $\hat{y} \in \{A, B, C, D, X\}$, recorded alongside the ground-truth answer and

a correctness flag in the structured JSON output.

3.4 Implementation

This section explains how the algorithmic design was implemented using specific software and hardware.

3.4.1 Module Implementation

Loader (`loader.py`)

The class `VideoLoader`, loads the Video-MME dataset [12] from HuggingFace using the library, `datasets` and transforms it to a `DataFrame` using the `pandas` library. The method `filter()` can accept optional arguments to filter for specific durations, domains, sub-categories, and task types. The method `get_videos()` filters the dataset, returns rows by URL, and collects all questions for a video into a dictionary, alongside the video ID, duration tier, and domain.

Preprocessor (`preprocessor.py`)

The class `Preprocessor`, downloads the videos and performs frame extraction. Videos are downloaded using the Python library `yt-dlp`, saving MP4 files from the URL at 720p resolution. To minimise YouTube’s bot-detection software, a known challenge in video research [26], the class uses exported browser session cookies and randomised sleep intervals of 8 to 20 seconds via the `time` library.

Frame extraction is executed through the OpenCV library. The native frame rate of the video is read and rounded to the nearest integer to handle non-integer frame rates (e.g., 29.97 fps). An extraction rate of one frame per second is used, and a frame is saved as `frame_XXXX.jpg`, where the four-digit index equals the timestamp in seconds. Both downloading and extraction contain checks to avoid re-downloading and/or reprocessing a video.

Indexer (`indexer.py`)

The class `Indexer` encodes frames with CLIP ViT-B/32 using the `sentence-transformers` library, processing them in batches of 32. The encoded frame embeddings are stacked into a float32 matrix, L2-normalised using `faiss.normalize_L2()`, and added to a `faiss.IndexFlatL2` index. The FAISS index and numPy array of frame paths are saved to disk. The method `already_indexed()` prevents re-encoding on subsequent runs.

Retriever (retriever.py)

The **Retriever** class implements both uniform sampling and semantic retrieval. For semantic retrieval, the question and options are concatenated, then encoded with CLIP, L2-normalised, and submitted to the FAISS index. The adaptive gap is computed as `max(30, total_frames // (k * 2))` and the greedy diversity filter is applied as described in Algorithm 1. If fewer than k frames are selected, a fallback fills remaining slots from the candidate set without the gap constraint. For uniform sampling, `numpy.linspace(0, total-1, k)` produces k evenly spaced indices including both endpoints.

Responder (responder.py)

This **Responder** class loads the LLaVA-Next 13B model (`llava-hf/llava-v1.6-vicuna-13b-hf`) [20]. In the single retrieval mode, only the top-ranked frame is passed to the model. The tiled retrieval mode sorts all k frames chronologically by filename index, resizes each to 336×336 pixels using Lanczos resampling, and arranges them into a 1008×672 black RGB canvas with a 3-column layout. Inference uses greedy decoding (`do_sample=False, max_new_tokens=10`). The decoded output is scanned for the first valid answer letter (A–D); if none is found, the prediction is recorded as **X**. The model is instantiated once and shared across both responder instances to avoid loading the 13-billion-parameter model twice.

Evaluator (evaluator.py)

The **Evaluator** class performs evaluation. For a video and its three questions, all three strategies are run in sequence: `clip_single`, `clip_tiled`, and `uniform`. The output result is recorded per question, including question ID, task type, video domain, ground-truth answer, retrieved frame paths, and a correctness flag for each retrieval strategy. The full output is saved as a structured JSON file named `video_id_runN.json`, where N increments to preserve results from previous evaluation runs.

3.4.2 Hardware and Infrastructure

All experiments under the pipeline are conducted on the University of Sheffield’s Stanage High Performance Computing (HPC) cluster. All jobs are allocated on a single NVIDIA A100 40GB GPU, providing sufficient memory to load LLaVA-Next 13B and process the frame tiling inputs without any memory overflow errors. Jobs are submitted through the SLURM workload manager scripts, with each job processing a batch of videos from the Video-MME dataset. The HPC infrastructure was required for this pipeline as the 13-billion parameter model could not be run on standard hardware.

3.4.3 Software Environment

The pipeline is implemented in Python 3.12.2 within a dedicated conda virtual environment. The key dependencies are listed in Table 3.2 and in the `requirements.txt` file.

Library	Version	Role
PyTorch	2.5.1+cu121	Deep learning framework for model inference
Transformers (HuggingFace)	5.2.0	LLaVA-Next model loading and inference
sentence-transformers	5.2.3	CLIP ViT-B/32 frame encoding
FAISS-GPU	1.9.0	Vector indexing and nearest-neighbour search
OpenCV	4.12.0	Video loading and frame extraction
yt-dlp	latest(2026.03.17)	YouTube video downloading
datasets (HuggingFace)	4.6.1	Video-MME dataset loading

Table 3.2: Software dependencies of the RAG pipeline.

3.5 Ethical and Legal Considerations

3.5.1 Data Usage

The Video-MME dataset [12] contains publicly available YouTube videos. All videos are accessed and downloaded through their public URLs using `yt-dlp` for research purposes only. No videos are redistributed or stored beyond the project’s duration. Once a video has been evaluated using the pipeline, the downloaded MP4 files are subsequently deleted. This is consistent with standard academic practice in the video understanding research field, where publicly available online videos are regularly used for benchmark evaluation [26] [10].

3.5.2 Model Usage

LLaVA-Next 13B is an open-source model available under a permissive research license and accessible through the Hugging Face model hub. The use of the model is consistent with the intended research application. Throughout the pipeline, no personally identifiable information is processed at any stage. The pipeline does not generate, store, or distribute any model outputs, except for the JSON files containing the predicted answer letters used for accuracy evaluation.

3.5.3 Environmental Considerations

Running the large-scale 13-billion parameter, LLaVA-Next model required an estimated 40 GPU-hours on an A100. To reduce unnecessary computation, the pipeline caches FAISS

indexes and downloaded videos to manage memory constraints and avoid redundant processing. Evaluation is batched via SLURM scripts to make efficient use of allocated GPU time on the HPC cluster.

3.6 Summary

This chapter has described the design and implementation of the retrieval-augmented pipeline proposed in this project. The pipeline is training-free, query-aware, and single-pass — addressing the three core limitations identified in the literature review. It consists of six modular components: a loader, preprocessor, indexer, retriever, responder, and evaluator. The central contribution is the semantic retrieval strategy, which uses CLIP embeddings and FAISS nearest-neighbour search with temporal windowing to retrieve the $k = 5$ frames most semantically relevant to a given question. This is compared against a uniform sampling baseline that selects frames without any reference to the query. Both methods pass their retrieved frames to LLaVA-Next 13B via a frame tiling strategy that respects the model’s context window constraint. The following chapter describes the experimental design used to evaluate these retrieval strategies on the Video-MME benchmark.

Chapter 4

Experiments

4.1 Overview

This chapter describes the experimental design used to evaluate the retrieval-augmented pipeline. It discusses the videos selected from the Video-MME benchmark, the three techniques being compared, the evaluation metric, and the dimensions used to analyse the results.

4.2 Benchmark Dataset: Video-MME

4.2.1 Why End-to-End Evaluation

Previous approaches considered using retrieval-only benchmarks such as YouCook2 [1] and ActivityNet-QA [49], which were evaluated using Recall@K and Median Rank. In short, these metrics measure whether the correct video segment appears in the top- k retrieved results. However, this goal does not match the main research question of this project.

The core research question of this project is not whether CLIP retrieval returns semantically relevant frames in isolation, but whether the returned frames yield more accurate answers from the downstream LLM than a uniform sampling baseline. This approach is consistent with the evaluation methodology used in VideoAgent [10] and SeViLA [48], both of which evaluate retrieval quality through downstream QA accuracy. Answering this question with Recall@K is ineffective because it measures retrieval performance in isolation. A pipeline with high Recall@K may perform poorly at question answering if the retrieved frames lack the visual details needed to answer the question.

Furthermore, a benchmark spanning multiple domains, question types, and video durations is necessary to identify the conditions under which semantic retrieval provides an advantage over uniform sampling. Therefore, end-to-end evaluation is appropriate, as it measures whether better frame selection produces higher accuracy values and directly addresses the core research question.

4.2.2 Video-MME

The chosen benchmark for the project is Video-MME [12]. The dataset contains 900 videos acquired from YouTube, covering 6 primary domains and 30 fine-grained subcategories. Each video contains 3 multiple-choice questions, yielding 2,700 question-answer pairs in total. The six domains are: Knowledge, Film & Television, Sports Competition, Life Record, Artistic Performance, and Multilingual as shown in Figure 4.1, providing coverage across various genres that differ substantially in visual information and reasoning requirements.



Figure 4.1: The six primary domains and 30 sub-categories of the Video-MME benchmark [12].

4.2.3 Duration Structure

The most important feature of the Video-MME benchmark, relevant to this project, is video duration, which is divided into three tiers. Short videos are under 2 minutes, with an average duration of 81 seconds. Medium videos range from 4 to 15 minutes, with an average duration of 520 seconds. Long videos range from 30 to 60 minutes, with an average duration of 2,741 seconds. Each domain contains approximately 50 videos per duration tier.

This choice is directly relevant to the research question. As video duration increases, the failure mode behind uniform sampling becomes more severe; for a 45-minute video with

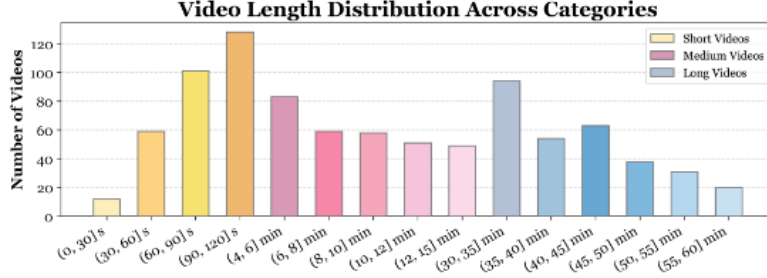


Figure 4.2: Video Duration Distribution: Video length distribution across categories [12]

$k = 5$ frames, each sampled frame represents a 9-minute window, and even a 10-second event has 93.8% probability of being missed entirely (see Section 2.3.3). If the core hypothesis holds, the performance advantage of semantic retrieval over uniform sampling should widen with video duration, as the failure mode of uniform sampling becomes more severe.

4.2.4 Question Types and Temporal Difficulty

Video-MME contains various question types across duration tiers. Short videos focus on perceptual tasks such as object recognition, action recognition, and attribute perception, where the answer is within a small number of frames. Long videos focus on temporal reasoning tasks that require evidence across temporally distant segments. These question types are shown in Figure 5.2

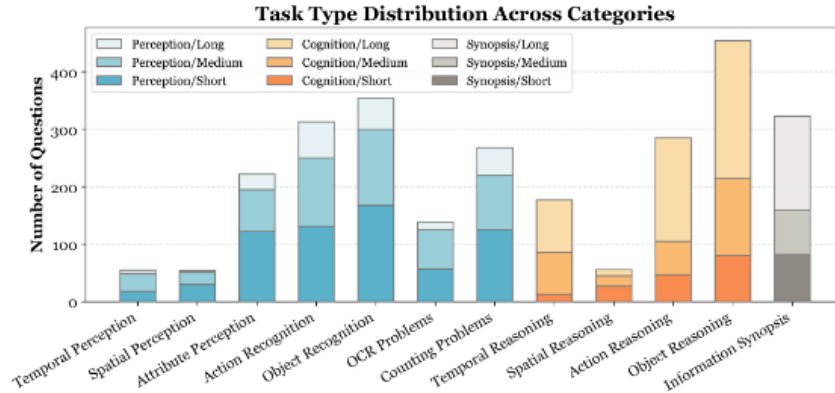


Figure 4.3: Task Type Distribution: Task type distribution across categories [12]

The benchmark introduces the use of certificate length, the minimum duration of a video that a human verifier needs to watch to confirm an annotation is correct. The median certificate length is 26.0 seconds for short videos, 167.3 seconds for medium videos, and 890.7 seconds for long videos [12]. For long videos, the relevant evidence spans a 15-minute window, and uniform sampling over a 45-minute video cannot reliably locate it using only $k = 5$ frames.

Semantic retrieval can target frames whose visual content is most aligned with the query, making it better suited to locate evidence within long videos, provided the relevant content is distinctive.

4.2.5 Alternative Benchmarks Considered

Multiple alternatives were considered before selecting Video-MME. EgoSchema [27] is a long-form benchmark used by VideoAgent [10] to evaluate training-free retrieval. However, its narrow egocentric domain and narrative questions disproportionately penalise visual-only retrieval systems, preventing a fair assessment of where CLIP-based retrieval succeeds or fails. NExT-QA [46] provides causal and temporal question answering, but its video duration averages under 2 minutes, presenting no context bottleneck. ActivityNet-QA [49] contains longer videos but uses open-ended questions evaluated with n-gram overlap metrics, such as BLEU and METEOR, which correlate poorly with semantic correctness [26].

Video-MME was selected because it combines duration tiers, a diverse range of domains, a variety of question types, unambiguous ground truth, and published baselines from prior systems. It is the most appropriate benchmark for evaluating the specific hypothesis addressed in this project.

4.3 Video Selection

4.3.1 Sample Construction

Evaluating the entire dataset requires downloading and processing all videos, which is infeasible given this project’s time constraints and computing budget. Therefore, a stratified sample was created to ensure all domains and durations were covered. The final sample consisted of 590 unique videos, totalling 1,770 multiple-choice questions. This is summarised in Table 4.1.

Duration Tier	Videos	Questions
Short (<2 min)	219	657
Medium (4 – 15 min)	165	495
Long (30 – 60 min)	206	618
Total	590	1,770

Table 4.1: Distribution of the evaluation sample by video duration tier.

4.3.2 Infrastructure

All pipeline jobs were submitted to the University of Sheffield’s Stanage High Performance Computing (HPC) cluster via the SLURM workload manager. Each job was allocated a single NVIDIA A100 40GB GPU and processed a batch of videos from the sample. Given that each video requires downloading, frame extraction, CLIP encoding, FAISS indexing, and

LLaVA-Next inference across three conditions, batching was essential to make efficient use of allocated GPU time. Job scripts specified memory, time limits, and output logging paths; failed jobs were identified from SLURM logs and resubmitted where possible.

4.3.3 Data Collection Challenges

The pipeline downloaded videos with the Python library `yt-dlp` from YouTube URLs from the Video-MME dataset. During extensive data collection, YouTube’s bot-detection procedures frequently blocked download requests, causing a subset of jobs to fail at the download stage, preventing frame extraction and further processing. This is a known challenge in video research and was mentioned by Maaz et al. [26], reflecting the difficulties in working with web-hosted datasets from platforms like YouTube.

This issue was partially resolved by exporting YouTube browser cookies to authenticate download requests, introducing random sleep intervals between downloads, and resubmitting failed jobs. Manually downloading failed videos was not a viable option due to the previously mentioned time constraints. Therefore, failed videos were excluded from the sample, which explains the gaps in the video IDs visible in the per-video results. It is acknowledged that download failures may not be uniformly distributed across duration tiers; longer videos are more susceptible to rate limiting, potentially introducing sampling bias in the long-duration results. This limitation is discussed further as a limitation in Chapter 6.

4.4 Experiment Conditions

Three techniques are evaluated for every video and question in the video sample. All techniques use LLaVA-Next 13B [20] as the downstream model and the same $k = 5$ frames per question, ensuring that any observed differences in accuracy are attributable solely to the retrieval strategy.

The three techniques are listed below:

1. **clip_single**: Semantic retrieval using CLIP with temporal windowing, as explained in Chapter 3. The top $k = 5$ frames are retrieved using CLIP with temporal windowing, but only the single highest-ranked frame is passed to LLaVA-Next.
2. **clip_tiled**: Identical to **clip_single**, but the top five frames are selected and concatenated into a single tiled grid image before being passed to the LLaVA-Next model (See Figure 3.2).
3. **uniform**: The baseline of this project. From a video, five frames are selected at evenly spaced intervals over the full duration of the video, without reference to the question. These five frames are concatenated into a single grid to match the format of **clip_tiled**.

The comparison between **clip_tiled** and **uniform** is the main experimental comparison, in which both techniques provide the model with five frames in the same tiled format, making the

retrieval strategy the sole variable. The comparison between `clip_single` and `clip_tiled` is a secondary comparison that analyses the additional benefit of multi-frame retrieval over single-frame retrieval.

4.5 Evaluation Metric and Analysis Dimensions

4.5.1 Evaluation Metric

The main metric used to evaluate multiple-choice answers is accuracy: the proportion of questions for which the predicted answer letter matches the ground-truth answer letter.

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Total number of questions}} \quad (4.1)$$

This metric was selected for several reasons. First, each question has a single unambiguous ground-truth answer. More importantly, accuracy has been used in published baselines and prior work (MovieChat [33], Video-ChatGPT [26]). Another reason is the near-uniform answer distribution in Video-MME [12], which is approximately 25.1% / 27.2% / 25.3% / 22.4% across the four options, implying that random guessing yields approximately 25% accuracy, thereby establishing a clear lower bound for accuracy.

4.5.2 Analysis Dimensions

Results are analysed along three dimensions in addition to overall accuracy. Each dimension addresses a specific segment of the research question.

1. **By duration tier** (short, medium, and long). This addresses Research Question 2: whether the accuracy of semantic retrieval under uniform sampling increases with video duration. This meets the prediction from the failure-mode analysis of uniform sampling in Section 2.3.3.
2. **By domain** (across all six Video-MME domains). This addresses Research Question 3: whether certain domain types favour semantic retrieval over uniform sampling.
3. **By task type** (across the 12 question categories defined in Video-MME). This reveals the specific reasoning capabilities that semantic retrieval enhances or fails to address, and is the most granular breakdown. It differentiates between perceptual tasks, such as object recognition, and temporal reasoning tasks. It also reveals where CLIP’s lack of temporal awareness limits retrieval performance.

4.6 Comparison against Published Baselines

An important requirement discussed in the literature review is that results must be compared against prior work rather than evaluated in isolation. However, it is important to note that direct comparison between this project and prior work is not the primary goal of this evaluation.

Published baselines differ from this project in terms of model size, training data, frame sampling strategy, and several other parameters. The more significant comparison is among `clip_single`, `clip_tiled`, and `uniform`, all of which use the same model, the same frame budget, and the same benchmark, thereby making the retrieval strategy the sole variable.

The Video-MME paper [12] reports results for several models evaluated under the same benchmark conditions, allowing for direct comparison. Table 4.2 presents the published results.

Model	Params	Short	Medium	Long	Overall
Random baseline	—	25.0%	25.0%	25.0%	25.0%
Video-LLaVA [22]	7B	—	—	—	39.9%
LLaVA-NeXT-Video [20]	7B	—	—	—	43.3%
LLaVA-NeXT-Video [20]	34B	63.1%	51.1%	44.6%	52.5%
Uniform (this project)	13B	49.0%	39.6%	36.2%	41.9%
CLIP Single (this project)	13B	53.4%	42.0%	38.3%	45.0%
CLIP Tiled (this project)	13B	54.3%	42.0%	38.8%	45.5%

Table 4.2: Comparison of this project’s results against published Video-MME baselines. Published results are taken directly from Fu et al. [12]. The uniform baseline uses the same LLaVA-Next 13B model as the CLIP conditions.

Two observations contextualise these results ahead of the full analysis in Chapter 5. The uniform baseline (41.9%) already exceeds Video-LLaVA 7B (39.9%) despite a fixed budget of only 5 frames, demonstrating that LLaVA-Next 13B provides a stronger reasoning backbone. CLIP Tiled (45.5%) exceeds LLaVA-NeXT-Video 7B (43.3%) using only 5 semantically selected frames, suggesting query-aware retrieval compensates for a restricted frame budget.

4.7 Summary

This chapter has explored the experimental design used to evaluate the retrieval pipeline. Video-MME was chosen as the benchmark dataset because it uniquely combines a diverse range of domains, question types, durations, and unambiguous ground truth, properties that are necessary to evaluate the hypothesis of this project.

The three techniques compared: `clip_single`, `clip_tiled`, and `uniform`, all use the same model and frame extraction budget. Accuracy is the primary evaluation metric used to analyse results by duration tier, domain, and task type, addressing all three research questions. The duration-stratified results in Chapter 5 represent the most analytically informative

component of the evaluation, as they directly test whether the performance advantage of semantic retrieval grows as the failure mode of uniform sampling becomes more severe.

Chapter 5

Results and Analysis

5.1 Overall Accuracy

This section presents the overall accuracy across the three retrieval strategies, summarised in Table 5.1.

Method	Correct	Overall Accuracy
Uniform Sampling	742/1770	41.9%
CLIP Single	796/1770	45.0%
CLIP Tiled	805/1770	45.5%

Table 5.1: Overall accuracy across all 1,770 evaluated questions.

Both semantic retrieval techniques outperform the uniform sampling baseline, showing that question-aware frame selection improves downstream reasoning accuracy. CLIP Tiled achieves the highest overall accuracy at 45.5%, followed closely by CLIP Single at 45.0%. Finally, the uniform sampling baseline achieves 41.9%, producing a 3.6% gap between CLIP Tiled and the baseline.

This finding is consistent with prior work and demonstrates that question-aware key-frame selection improves accuracy. SeViLA [48] demonstrates this on the NExT-QA benchmark (63.6%), outperforming the uniform sampling baseline (59.7%) by 3.9%. The zero-shot results were chosen for SeViLA as this project does not rely on any fine-tuning or pre-training.

To judge whether the advantage is consistent per-video, Table 5.2 reports win, tie, and loss counts for each retrieval strategy against the baseline across all 590 evaluated videos.

As shown in Table 5.2, CLIP Single wins on 160 videos and loses on 124, producing a win rate of 56.3%. CLIP Tiled wins on 148 videos but loses on only 100 videos, producing a win rate of 59.7%. There are many ties, reflecting the important fact that each video contains three questions. When both methods answer the same questions correctly, a tie is recorded. These values confirm that the accuracy advantage is not due to a small number of outlier videos but rather reflects a consistent per-video advantage in semantic retrieval.

Method	Wins	Ties	Losses	Win Rate
CLIP Single	160	306	124	56.3%
CLIP Tiled	148	342	100	59.7%

Table 5.2: Per-video win, tie, and loss counts for each semantic retrieval condition versus the uniform baseline across 590 videos. Win rate is computed over decisive outcomes only (wins + losses). A win indicates the semantic condition answered more questions correctly on that video.

The small difference between CLIP Tiled and CLIP Single is notable. Although CLIP Tiled provides the top five retrieved frames rather than the top one, the improvement is only 0.5%. This suggests that increasing the number of frames retrieved does not necessarily translate into noticeable accuracy gains. One reason is the reduction in spatial resolution due to resizing. LLaVA-Next’s [20] visual encoder processes the composite grid image from the top-five ranked frames as a single input at fixed resolution, reducing the spatial detail available to the model compared to passing a single full-resolution frame.

The uniform sampling baseline performance of 41.9% is comparatively strong, indicating that the questions within Video-MME [12] remain answerable through sparse uniformly sampled evidence. This result exceeds Video-LLaVA 7B [22] (39.9%) overall on the full Video-MME benchmark despite processing fewer frames (5 vs 8), and suggests a stronger reasoning model compensates for the reduced frame budget. Consequently, the main advantage of retrieval is that it increases the likelihood that the chosen frames contain visual evidence directly relevant to the query. This principle is established for text-based RAG systems by Lewis et al. [19], who illustrate that retrieval quality is the driving factor for downstream answer quality, and is extended here to visual retrieval.

5.2 Results by Duration Tier

This section presents accuracy divided by duration, the primary analysis dimension that addresses Research Question 2: whether the performance advantage of semantic retrieval over uniform sampling grows as video duration increases.

Method	Short	Medium	Long	Overall
Uniform Baseline	49.0%	39.6%	36.2%	41.9%
CLIP Single	53.4%	42.0%	38.3%	45.0%
CLIP Tiled	54.3%	42.0%	38.8%	45.5%

Table 5.3: Overall and duration-stratified accuracy for all three experimental conditions across 1,770 questions.

Figure 5.1 and Table 5.3 present accuracy values categorised by duration tier. Three key patterns are evident.

First, semantic retrieval outperforms uniform sampling across all three duration tiers. This

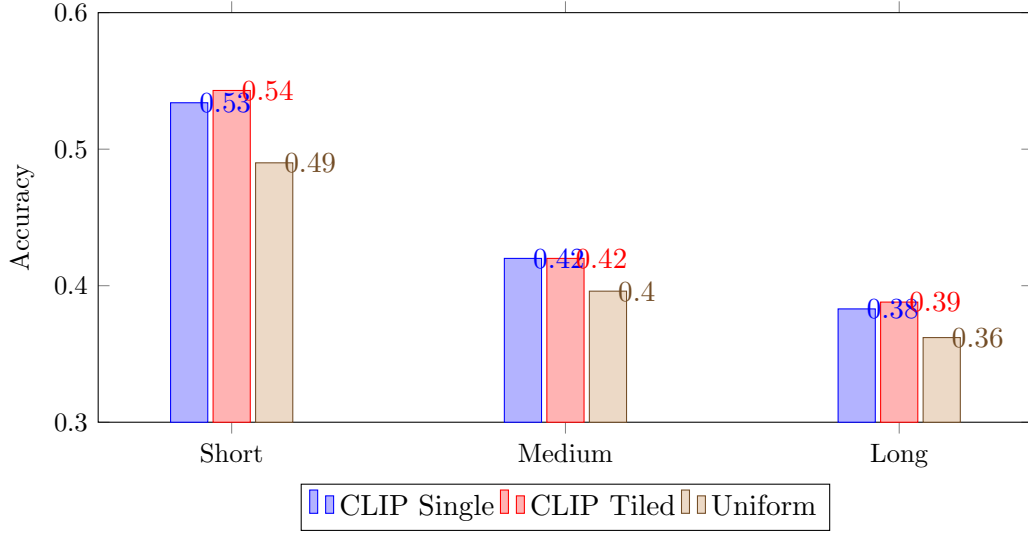


Figure 5.1: Accuracy by duration tier for all three experimental conditions. The y-axis begins at 30% to make differences between methods visible. The random guessing baseline is 25%.

advantage confirms that question-aware frame selection consistently yields better visual evidence than question-agnostic sampling, regardless of video length, thereby addressing Research Question 1 across all conditions.

Second, accuracy degrades with duration across all three retrieval techniques. The difference in short-video accuracy (49% to 54.3%) is larger than that in long-video accuracy (36.2% to 38.8%). This trend is expected because longer videos contain more visual content, making it more difficult to select the most semantically relevant frames for a given question. This pattern is consistent with published baselines across all evaluated models in the Video-MME benchmark paper [12], where Fu et al. report that accuracy degrades as duration increases, regardless of model architecture or scale.

However, contrary to the original hypothesis, the improvement gap between semantic retrieval and uniform sampling does not increase with video duration. The largest gap between semantic retrieval and uniform sampling is observed on short videos, where CLIP Tiled outperforms Uniform by 5.3%. This gap degrades to 2.4% for medium videos, and 2.6% for long videos. The advantage of short videos likely stems from the nature of short-video questions, which focus on perceptual tasks such as object recognition and attribute identification. In contrast, longer videos involve temporal reasoning tasks, for which CLIP’s static encoder provides no temporal context [30]. Furthermore, this result is consistent with the shift in question types across duration tiers as addressed by Fu et al. [12], who report that longer videos contain more temporal reasoning tasks. This can be shown by the distribution chart of task types (see Figure 5.2).

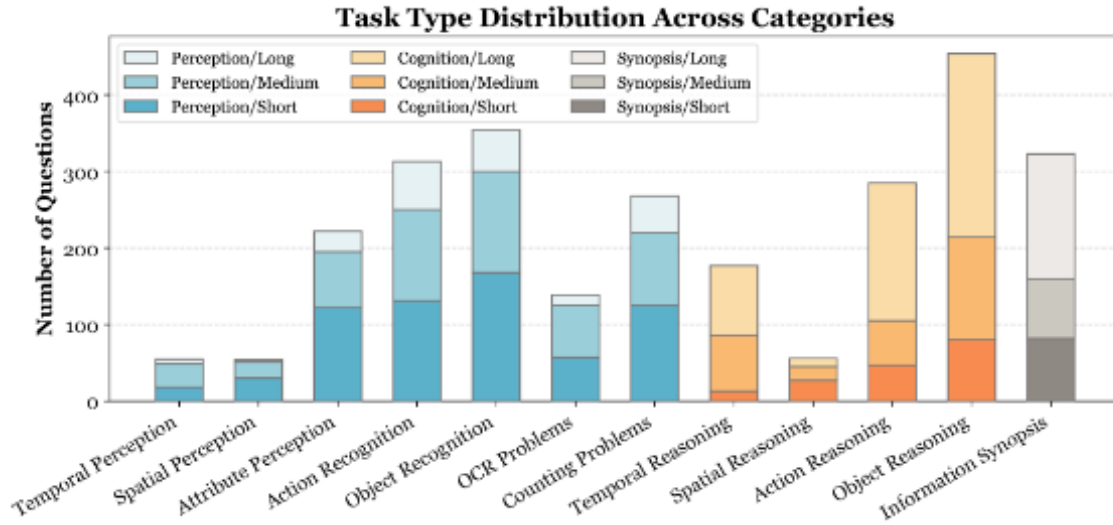


Figure 5.2: Distribution of question types across duration tiers in Video-MME [12]. Longer videos contain a higher proportion of temporal reasoning tasks, which directly affect the pattern of results observed in Section 5.2.

5.3 Results by Domain

Table 5.4 illustrates accuracy across the six Video-MME domains for all three retrieval strategies.

Domain	N	CLIP Single	CLIP Tiled	Uniform
Film & Television	306	48.0%	50.0%	43.8%
Knowledge	492	47.0%	46.1%	42.5%
Life Record	453	44.8%	47.2%	43.7%
Sports Competition	375	41.9%	42.7%	39.7%
Artistic Performance	108	40.7%	37.0%	37.0%
Multilingual	36	38.9%	30.6%	33.3%

Table 5.4: Accuracy by Video-MME domain for all three experimental conditions, sorted by CLIP Tiled accuracy.

Semantic retrieval outperforms the uniform baseline in four of the six domains. The largest gains are in Film & Television (CLIP Tiled +6.2 percentage points over uniform) and Knowledge (CLIP Single +4.5 percentage points). These domains contain questions that require finding a specific visual detail within a longer video. This is precisely the scenario where question-aware retrieval provides the greatest advantage over fixed-interval sampling. Two domains within the benchmark show uniform sampling that matches or outperforms semantic retrieval. In the Artistic Performance domain, CLIP Tiled and the uniform baseline

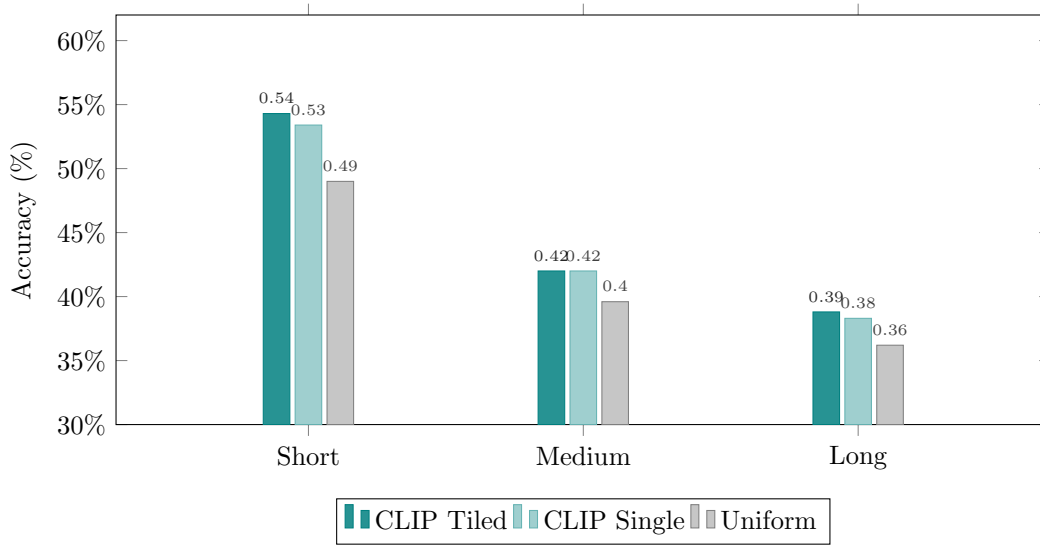


Figure 5.3: Bar chart comparing retrieval accuracy across the six Video-MME domains.

score are equal at 37.0%, with CLIP Single performing somewhat better at 40.7%. This result suggests that for content with dense, continuous information, such as performances in which frames share a similar structure, CLIP’s embeddings yield less discriminative frames, whereas uniform sampling provides better coverage.

In the Multilingual domain, the result is based on only 36 questions and should be interpreted with caution, given the small sample size. CLIP Tiled achieves the lowest accuracy at 30.6% against uniform (33.3%), and CLIP Single achieves the highest score at 38.9%. Multilingual questions involve on-screen text, and resizing frames to 336×336 pixels may reduce text readability, impairing the model’s ability to read the visual content and answer the question. Radford et al. [30] note that CLIP’s text understanding is derived from descriptive image captions rather than OCR tasks.

5.4 Results by Question Type

Table 5.5 illustrates the accuracy across the 12 question types defined in Video-MME. This breakdown provides the most granular evidence to explain the patterns observed across the results and identifies where semantic retrieval outperforms and underperforms the baseline. Figure 5.2 shows the distribution of question types across duration tiers in Video-MME, illustrating why longer videos are harder for all methods.

Question Types Where Semantic Retrieval is Most Effective

Attribute Perception demonstrates the largest advantage, where CLIP Single achieves 60.4% against a uniform baseline of 52.3%; an 8.1% gain. For Object Reasoning, CLIP Single

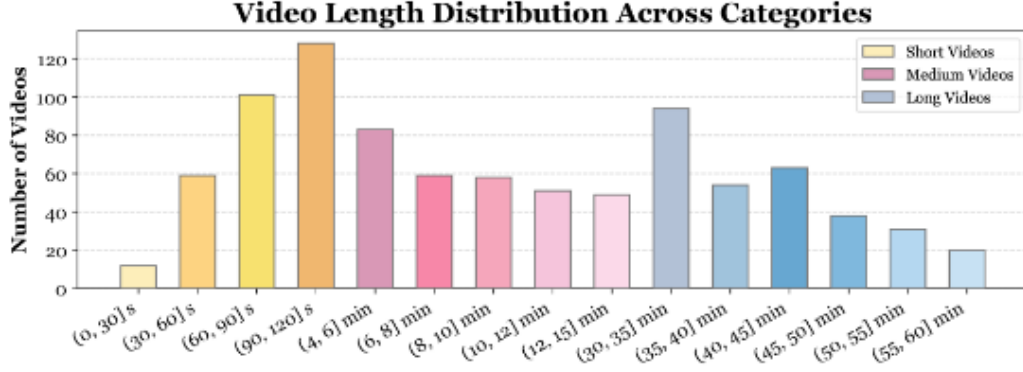


Figure 5.4: Distribution of video durations across the six Video-MME domains [12]. Each domain contains videos across short, medium, and long tiers.

Question Type	N	CLIP-S	CLIP-T	Uniform
Attribute Perception	149	60.4%	59.1%	52.3%
Spatial Reasoning	29	58.6%	51.7%	65.5%
Object Recognition	229	52.0%	51.1%	48.0%
Information Synopsis	207	51.2%	59.4%	54.1%
Spatial Perception	41	51.2%	56.1%	51.2%
OCR Problems	91	50.5%	48.4%	46.2%
Object Reasoning	298	46.3%	43.3%	39.9%
Temporal Perception	38	36.8%	44.7%	34.2%
Temporal Reasoning	107	40.2%	37.4%	35.5%
Action Recognition	181	43.6%	47.5%	42.0%
Action Reasoning	207	33.3%	33.8%	34.3%
Counting Problem	193	28.0%	27.5%	22.3%

Table 5.5: Accuracy by question type for all three conditions, sorted by CLIP Tiled accuracy. N denotes the number of questions of each type in the evaluated sample.

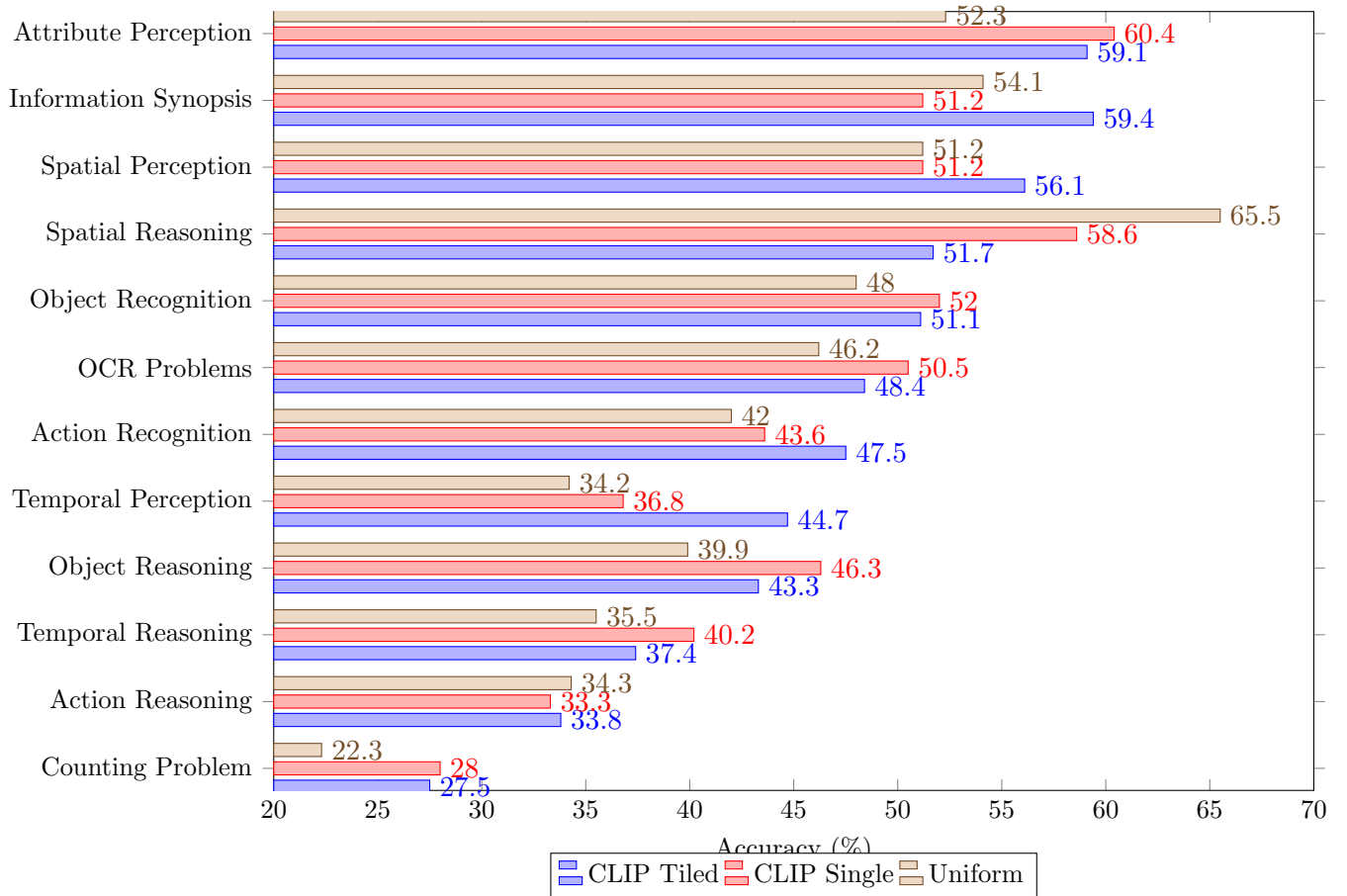


Figure 5.5: Accuracy across Video-MME question types for all three retrieval strategies.

achieves a 6.4% advantage over the uniform baseline. Object Recognition questions (CLIP Single +4.0pp) also show consistent and substantial gains. These question types require identifying a specific visual property, such as colour, texture, or an object, within the video, which directly aligns with CLIP’s retrieval of matching visual content from text [30]. Question-aware retrieval locates the frame containing the relevant object or attribute, while uniform sampling may not.

5.4.1 Question Types Where Semantic Retrieval Underperforms

Three question types highlight marginal results and require analysis. For Action Reasoning questions, CLIP Single (33.3%) and CLIP Tiled (33.8%) are both below the uniform baseline (34.3%). This is consistent with the known limitation of CLIP highlighted in Section 2.5.3. CLIP encodes static frames and does not represent motion, direction, or relationships between frames [30]. A particular question asking about an action requires understanding a temporal sequence, not just identifying static frames. This finding matches the results of Luo et al. [25], who note that while frame-level CLIP embeddings are sufficient for visual-based retrieval, they discard the temporal relationships between frames that action understanding requires.

Spatial Reasoning shows similar results, with the baseline (65.5%) outperforming both CLIP Single (58.6%) and CLIP Tiled (51.7%). However, this category covers only 29 questions, and the results should be interpreted cautiously given the small sample size, as a single misclassified question corresponds to a 3.4% change.

5.4.2 Information Synopsis

This is the most noteworthy finding for CLIP Tiled, specifically as it achieves 59.4% against the uniform baseline of 54.1%; a 5.3% gain, while CLIP Single achieves only 51.2%, 2.9% below the baseline. This suggests that for summary-related questions, the multi-frame, tiled format provides the model with a broader temporal snapshot of the video, improving comprehension. By contrast, the single-frame approach provides a narrower snapshot and is therefore less effective than the tiled approach.

5.4.3 Counting Problems

These questions are the hardest category across all methods (27.5% to 28.0% for CLIP Single/Tiled, 22.3% for uniform), all close to the 25% random-guessing baseline. These types of questions require counting objects or events across multiple frames, a feat that neither the retrieval strategy nor the model’s reasoning addresses effectively at this frame budget. This limitation reflects a broader challenge for Vid-LLMs documented across multiple benchmarks as documented by Fu et al. [12]

5.5 Comparison against Existing Approaches

Comparing the results of this project against published Vid-LLM baselines requires caution, as published models differ greatly from this project in model size, training data, number of input frames, and inference strategy. Therefore, the results cannot be considered a fair comparison with other models. However, placing the results within the published surroundings offers a useful way to interpret their importance.

Table 4.2 from Chapter 4 illustrates the full range of published baselines. Two observations are relevant here. There are three main observations that contextualise these results.

Model	Params	Short	Medium	Long	Overall
Random baseline	—	25.0%	25.0%	25.0%	25.0%
Video-LLaVA [22]	7B	45.3%	38.0%	36.2%	39.9%
LLaVA-NeXT-Video [20]	7B	—	—	—	43.3%
LLaVA-NeXT-Video [20]	34B	63.1%	51.1%	44.6%	52.5%
Uniform (this project)	13B	49.0%	39.6%	36.2%	41.9%
CLIP Single (this project)	13B	53.4%	42.0%	38.3%	45.0%
CLIP Tiled (this project)	13B	54.3%	42.0%	38.8%	45.5%

Table 5.6: This project’s results alongside published Video-MME baselines. Published figures without subtitles are used throughout for consistency with this project’s evaluation conditions. Results for this project are evaluated on a 590-video stratified sample rather than the full 900-video benchmark.

First, the uniform sampling baseline (41.9%) exceeds Video-LLaVA 7B (39.9%) [22], despite this pipeline processing only 5 frames versus LLaVA’s 8. This confirms that LLaVA-NeXT 13B provides a stronger reasoning backbone. Furthermore, it illustrates that model quality and frame selection strategy both contribute to final accuracy, and that processing more frames does not produce stronger results.

Second, CLIP Tiled (45.5%) exceeds LLaVA-NeXT-Video 7B (43.3%) [20] despite using only 5 semantically selected frames. This shows that a training-free, single-pass retrieval pipeline with a low frame budget outperforms a purpose-built video language model. This supports the claim that question-aware frame selection compensates for a low frame budget without requiring fine-tuning.

Third, the gap between this pipeline and LLaVA-NeXT-Video 34B (52.5%) highlights the impact of model size on the retrieval strategy. A 34-billion-parameter model has more powerful reasoning capabilities than a 13-billion-parameter model. This gap points to a clear direction for future work: applying the retrieval strategy from this pipeline to larger, more powerful models.

5.6 Discussion

5.6.1 Why Semantic Retrieval is effective?

The results discussed throughout this chapter confirm that question-aware frame selection provides more relevant visual evidence than question-agnostic uniform sampling. The grounding for this is established in Section 2.3.3, where the probability of uniform sampling missing an important frame is 93.8% for a 10-second event within a 45-minute video. Semantic retrieval tackles this by encoding the query as a text embedding and then using it to retrieve the most semantically relevant frames.

This finding extends the principle established in text-based RAG systems [19] to the visual domain. The quality of the retrieved data is the primary determinant of the quality of the downstream answer. The question types in which this advantage is largest (Attribute Perception, Object Reasoning, Object Recognition) are visual tasks in which a specific detail must be located in the video. This is where CLIP’s contrastive pre-training makes it effective at matching text to visual content [30].

The question types from the Video-MME benchmark highlight where the advantage is the largest. For spatio-related questions in which a specific visual detail must be located in the video, CLIP’s pre-training on 400 million image-text pairs makes it effective at matching text to visual content, which the retrieval mechanism directly exploits.

5.6.2 Why the gap does not grow with duration

The results relating to duration do not show an increasing advantage for semantic retrieval. The gap for short videos is the largest (+5.3% for CLIP Tiled), decreasing for medium (+2.4%) and long videos (+2.6%). There are two plausible explanations.

First, the distribution of question types changes with duration. Questions for short videos are dominated by perceptual tasks, for which CLIP-based retrieval is more effective. Long videos contain more temporal reasoning and action questions, for which semantic retrieval provides no advantage or performs below baseline. This is illustrated by Figure 5.2 documented by Fu et al. [12]. This shift reduces the advantage of longer videos, even if retrieval helps with perceptual questions in long videos.

Second, CLIP lacks temporal awareness, and this limitation becomes more significant as video duration increases. For long videos with temporally demanding questions, the relevant information may span multiple non-contiguous frames that must be understood in sequence. Fu et al. [12] report a median certificate length of 890.7 seconds for long videos, meaning that relevant evidence spans 15 minutes on average. This highlights a fundamental ceiling on performance at the $k = 5$ frame budget, especially for long videos and explains why the gap does not widen.

5.6.3 Why CLIP Tiled and CLIP Single are similar

The 0.5% gap between CLIP Single (45.0%) and CLIP Tiled (45.5%) is smaller than expected. One potential reason is that LLaVA-Next was not trained on the tiled grid format used in this project. As mentioned before, frames are resized to 336×336 pixels and arranged in a 3×2 grid, but the encoder processes this as a single image. Li et al. [20] describe LLaVA-Next’s high-resolution tiling as a structured input format where tiles represent different segments of the same image, not different temporal moments from other frames. Therefore, the model cannot interpret spatial relationships between grid cells as temporal relationships.

The Information Synopsis result, in which CLIP Tiled outperforms CLIP Single by 8.2%, suggests that the tiled format provides additional benefit for summary-related questions. However, for most question types, the top-ranked single frame captures enough visual evidence.

5.6.4 Limitations

There are several limitations that constrain the conclusions that can be identified from these results.

YouTube bot-detection reduced the sample coverage from 900 videos to 590 videos. While the sample covers all duration tiers and domains, certain question types and domains had fewer videos and were therefore under-represented in the results.

All conditions use the same frame budget of $k = 5$ with no evaluation across different values of k . A larger frame budget could have potentially revealed new patterns or a different operating point for the tiled format. Further inferences could have been drawn from the results regarding the pipeline and its impact on the quality of the results.

As established in Section 2.5 and confirmed by the results, CLIP’s static image encoding has a negative effect on performance for temporally related question types. A temporally aware encoder, such as a video-specific model, could address this limitation, but at the cost of the training-free constraint.

LLaVA-Next was not trained on the specific tiled composite format used in this project. The small margin between CLIP Tiled and CLIP Single suggests that the model does not fully utilise the additional frames given by tiling, most likely due to the lack of interpretation of the grid layout as a temporal sequence.

Chapter 6

Conclusions and Future Work

6.1 Summary of the Project

This project investigated whether a training-free, single-pass, question-aware frame retrieval pipeline could outperform the standard uniform sampling baseline for multiple-choice question-answering on long-form videos. The motivation behind this project was a structural gap identified in the existing literature alongside the context-bottleneck problem within Vid-LLMs. There were no existing approaches that method simultaneously satisfied all three requirements. As identified in Chapter 2, these three properties are necessary for large-scale video evaluation without task-specific supervision.

The proposed CLIP-RAG pipeline addressed this gap by encoding each video frame as a 512-dimensional CLIP embedding, storing embeddings in a FAISS index, and at query time, retrieving the $k = 5$ frames most semantically similar to the concatenated question and answer options. An adaptive temporal windowing mechanism was designed and implemented as part of this project to prevent retrieved frames from clustering around a single temporal moment, enforcing diversity across the retrieved set. Retrieved frames were passed to LLaVA-Next 13B either as a single top-ranked frame (`clip_single`) or as a composite tiled grid (`clip_tiled`) for downstream answer generation. A uniform sampling baseline (`uniform`) selected frames at fixed intervals with no reference to the query and formatted as a composite tiled grid.

The pipeline was evaluated on a sample of 590 videos and 1,770 questions from the Video-MME benchmark [12], covering all three duration tiers and all six domains.

6.2 Conclusions

6.2.1 Primary Hypothesis

The primary hypothesis is confirmed. CLIP-based semantic retrieval outperforms uniform sampling overall across all 1,770 evaluated questions. CLIP Tiled achieves 45.5% accuracy and CLIP Single achieves 45.0%, compared to 41.9% for the uniform baseline, producing a

3.6 percentage point improvement for the best-performing condition. The per-video win rate analysis confirms this is a consistent advantage rather than a result driven by a small number of outlier videos, with CLIP Tiled winning on 59.7% of decisive video comparisons against the baseline.

6.2.2 Research Question 2: Duration

The prediction that the performance advantage would grow with video duration was not fully supported by the results. The largest absolute gap between CLIP Tiled and uniform sampling is observed for short videos (+5.3 percentage points), with smaller gaps for medium (+2.4pp) and long (+2.6pp) videos. Two factors explain this pattern.

First, the distribution of question types shifts with duration as documented by Fu et al. [12]. Shorter videos contain more perceptual tasks such as object recognition and attribute identification, where CLIP’s visual-semantic alignment provides the greatest benefit. Longer videos introduce more temporal reasoning and action reasoning tasks, where CLIP’s static image encoding provides no advantage, as established in Section 2.5.3.

Second, the certificate length data from Fu et al. [12] shows that the relevant evidence for long video questions span approximately 15 minutes on average. No 5-frame retrieval system can reliably cover this window regardless of selection strategy, establishing a fundamental limit on performance at this frame budget that affects all three conditions equally.

6.2.3 Research Question 3: Domain and Question Type

Semantic retrieval provides the largest gains in Film & Television (+6.2pp for CLIP Tiled) and Knowledge (+4.5pp for CLIP Single). These are domains where questions require locating fine-grained visual detail within a longer video, which is precisely the scenario where question-aware retrieval provides the greatest advantage over uniform sampling, as discussed in Section 5.3.

The advantage is absent or reversed in the Multilingual and Artistic Performance domains. In Artistic Performance, dense visual homogeneity reduces CLIP’s ability to distinguish between frames. In the Multilingual domain, on-screen text is degraded by the frame resizing required for the tiled input format, reducing the model’s ability to read content that is directly relevant to the question.

At the question-type level, the largest gains are on Attribute Perception (+8.1pp), Object Reasoning (+6.4pp), and Object Recognition (+4.0pp). These are tasks that require identifying a specific visual property, directly aligning with CLIP’s contrastive pre-training objective. Action Reasoning and Spatial Reasoning show reversals where uniform sampling outperforms both CLIP conditions, confirming the known limitation of CLIP discussed in Section 2.5.3: CLIP encodes static frame appearance and discards the temporal and causal relationships between frames that these question types require [30].

6.2.4 Contextualisation Against Prior Work

Placing these results in the context of published baselines provides further perspective. CLIP Tiled (45.5%) exceeds LLaVA-NeXT-Video 7B (43.3%) [20] on the Video-MME benchmark despite using only 5 semantically selected frames. This demonstrates that question-aware frame selection compensates for a restricted frame budget without any task-specific training. The uniform baseline (41.9%) also exceeds Video-LLaVA 7B (39.9%) [22] despite processing fewer frames, suggesting that the stronger LLaVA-Next 13B reasoning backbone contributes independently of retrieval strategy. The gap between this project and LLaVA-NeXT-Video 34B (52.5%) reflects the difference in model scale rather than retrieval design, and represents a direction for future work.

6.3 Future Work

6.3.1 Using other modalities

This project only uses the visual information from a video in the form of frames. Including the extraction of relevant audio and captions can improve accuracy scores, especially across questions testing temporal functionality.

6.3.2 Larger Backbone Models

The most direct extension of this work is to apply the CLIP-RAG retrieval strategy to larger and more capable downstream models. The gap between CLIP Tiled and LLaVA-NeXT-Video 34B reflects reasoning capacity rather than retrieval quality. Applying semantic retrieval to models such as LLaVA-NeXT-Video 34B or Qwen2-VL 72B would determine whether the retrieval advantage transfers to stronger reasoning systems and whether the performance gap over uniform sampling widens at larger model scale.

6.3.3 Temporally-Aware Retrieval Encoders

The results demonstrate that CLIP’s static image encoding is the primary limitation for temporal and action reasoning question types. A natural extension is to replace CLIP with a temporally-aware encoder that can represent motion, causality, and event sequences within its embedding space. This would directly address the failure mode identified in the question-type results while potentially retaining the training-free property if a suitable pre-trained encoder is available.

6.3.4 Frame Budget Ablation

All three conditions in this project use $k = 5$ frames with no evaluation across different values of k . Evaluating performance at a range of frame budgets would reveal the relationship

between frame count and accuracy for both retrieval strategies, and determine whether the advantage of semantic retrieval over uniform sampling grows or shrinks as more frames are provided to the model.

6.3.5 Runtime and Efficiency Analysis

A formal per-query latency comparison between semantic retrieval and uniform sampling was not conducted due to constraints on fine-grained logging during HPC batch evaluation. Future work should measure wall-clock time per video across all pipeline stages, providing a rigorous comparison of computational cost between the single-pass design of this pipeline and iterative approaches such as VideoAgent [10].

6.3.6 Full Dataset Coverage

YouTube bot-detection limited the evaluation sample to 590 of the 900 available Video-MME videos. Evaluating on the full benchmark would increase the statistical reliability of the results, particularly for smaller sub-categories such as Multilingual, which contained only 36 questions in this evaluation and produced results that should be interpreted cautiously given the small sample size.

6.4 Personal Reflection

This project provided practical experience that extends beyond typical coursework. Implementing and evaluating a 13-billion parameter model required working at a scale that was previously unfamiliar, including submitting SLURM batch jobs to the Stanage HPC cluster, managing GPU memory constraints, and debugging pipeline failures from job logs rather than interactive terminals. These challenges reflect the conditions of real machine learning research and provided skills that coursework assignments do not typically introduce.

The results also provided a genuine research lesson. The hypothesis that the retrieval advantage would grow with video duration was grounded in the failure mode analysis of uniform sampling presented in Section 2.3.3, and it was partially incorrect. Understanding why it was incorrect, through the question-type distribution analysis and the certificate length argument, was more analytically valuable than a straightforward confirmation would have been. Interpreting unexpected results rigorously rather than dismissing them is something this project required directly and is a skill that will be useful going forward.

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Appendices

Appendix A: Software Environment

The pipeline was implemented in Python 3.12.12 within a dedicated conda virtual environment on the University of Sheffield’s Stanage HPC cluster. Table 1 lists the key dependencies used.

Package	Version
Python	3.12.12
PyTorch	2.5.1+cu121
transformers (HuggingFace)	5.2.0
sentence-transformers	5.2.3
faiss-gpu	1.9.0
opencv (py-opencv)	4.12.0
yt-dlp	2026.3.17
datasets (HuggingFace)	4.6.1
accelerate	1.12.0
numpy	1.26.4
pandas	2.3.3
pillow	10.4.0
tokenizers	0.22.2
safetensors	0.7.0
huggingface_hub	1.5.0
torchvision	0.20.1+cu121
tqdm	4.67.3
scikit-learn	1.8.0

Table 1: Key software dependencies of the pipeline. Full environment specification available in the submitted `requirements.txt` file.

The full `conda list` output, including all transitive dependencies and build strings, is provided in the submitted `requirements.txt` file accompanying this dissertation.

Appendix B: SLURM Batch Job Script

All evaluation jobs were submitted to the Stanage HPC cluster via the SLURM workload manager. The script below is representative of the job scripts used throughout the evaluation. Each job processed a list of video IDs sequentially, skipping any video already processed in the target output directory.

```
#!/bin/bash
#SBATCH --job-name=videorag_batch
#SBATCH --partition=gpu
#SBATCH --qos=gpu
#SBATCH --gres=gpu:1
#SBATCH --cpus-per-task=4
#SBATCH --mem=60G
#SBATCH --time=08:00:00
#SBATCH --output=logs/batch_%j.out
#SBATCH --error=logs/batch_%j.err

# Environment setup
module load Anaconda3/2024.02-1
source $(conda info --base)/etc/profile.d/conda.sh
conda activate videorag

export HF_HOME=/mnt/parscratch/users/$USER/hf_cache
export TRANSFORMERS_CACHE=/mnt/parscratch/users/$USER/hf_cache
mkdir -p $HF_HOME

cd ~/dissertation
mkdir -p logs
mkdir -p "$OUTPUT_DIR"

# Loop over video IDs
```



```

for VIDEO_ID in "${VIDEO_IDS[@]}"; do

    # Skip if already completed
    EXISTING=$(ls ${OUTPUT_DIR}/${VIDEO_ID}_run*.json \
        2>/dev/null | wc -l)
    if [ "$EXISTING" -gt 0 ]; then
        echo "[SKIP] Video $VIDEO_ID already done"
        continue
    fi

    python run_eval.py "$VIDEO_ID" "$OUTPUT_DIR"

    if [ $? -eq 0 ]; then
        echo "[SUCCESS] Video $VIDEO_ID completed"
    else
        echo "[FAILED] Video $VIDEO_ID failed"
    fi

done

```

Each job was allocated a single NVIDIA A100 40GB GPU with 60GB of system memory and an 8-hour time limit. The `HF_HOME` and `TRANSFORMERS_CACHE` variables redirect HuggingFace model downloads to the Stanage scratch filesystem, which has higher storage capacity than the home directory. The skip check prevents redundant reprocessing when jobs are resubmitted after partial failures.

Appendix C: Full Per-Video Results

Complete per-video and per-question results for all 590 evaluated videos are provided in the accompanying file `full_results.csv`. Each row contains the video ID, duration tier, domain, question ID, task type, ground-truth answer, and predicted answer letters for all three conditions (`clip_single`, `clip_tiled`, `uniform`), alongside a correctness flag for each.